PREDICTED PAPER 01 – ANSWER KEY & SOLUTIONS

Open only after the paper is done. Every answer was checked by the writer's verification script, an independent blind solver and a judge who re-derived the key and every distractor.

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	(2)	16	(4)	31	(3)	46	5	61	(4)
2	(4)	17	(2)	32	(2)	47	50	62	(3)
3	(1)	18	(1)	33	(4)	48	16	63	(1)
4	(3)	19	(3)	34	(1)	49	300	64	(2)
5	(2)	20	(4)	35	(2)	50	61	65	(4)
6	(1)	21	30	36	(3)	51	(2)	66	(3)
7	(3)	22	13440	37	(4)	52	(1)	67	(1)
8	(4)	23	10385	38	(2)	53	(4)	68	(4)
9	(1)	24	27	39	(1)	54	(3)	69	(2)
10	(2)	25	7	40	(3)	55	(2)	70	(3)
11	(4)	26	(3)	41	(4)	56	(4)	71	39
12	(3)	27	(1)	42	(1)	57	(1)	72	175
13	(1)	28	(2)	43	(2)	58	(3)	73	547
14	(2)	29	(4)	44	(4)	59	(2)	74	9
15	(3)	30	(1)	45	(3)	60	(1)	75	50

MATHEMATICS – SOLUTIONS

Q1. Answer: (2)

Hyperbola — Medium

Stem: Let the foci of a hyperbola H be $S(-1,3)$ and $S'(-1,-1)$, and let H pass through the point $P(11,8)$. If l is the length of the latus rectum of H and d is the distance of P from the directrix of H corresponding to the focus S, then $l+2d$ is equal to :

1. Dono foci line $x=-1$ par hain, so transverse axis vertical hai; centre = midpoint $(-1,1)$ aur $SS'=4 \Rightarrow c=2$.
2. $SP=\sqrt{12^2+5^2}=13$, $S'P=\sqrt{12^2+9^2}=15$; hyperbola par $|SP-S'P|=2a \Rightarrow a=1$.
3. $b^2=c^2-a^2=3$, $e=c/a=2$, latus rectum $l=2b^2/a=6$.
4. Focus $S(-1,3)$ (centre ke upar) ki directrix $y=1+a/e=3/2$; $d=8-3/2=13/2$ (check: $SP=e\,d=13$).
5. $l+2d=6+13=19$.

Final Answer: (2) 19

Trap: (1) 16 = latus rectum ko b^2/a (aadha) le liya; (3) 21 = doosre focus S' wali directrix se distance 15/2 use ki; (4) 23 = ellipse wala $b^2=a^2+c^2$ laga kar $l=10$ nikala.

Concept to recognise: Two foci on one vertical line plus a point on the curve: a comes from |SP - S'P| = 2a (vertical transverse axis), then b^2 = c^2 - a^2; 'directrix corresponding to S' signals the focus-directrix relation SP = e.d. *(Hyperbola parameter bookkeeping)*

Q2. Answer: (4)

Sequences & Series — Medium

Stem: The sum of the series $\frac{1^3}{2} + \frac{1^3+2^3}{2+4} + \frac{1^3+2^3+3^3}{2+4+6} + \cdots + \frac{1^3+2^3+\cdots+15^3}{2+4+\cdots+30}$ is equal to :

1. n-th term: $T_n = \dfrac{1^3+2^3+\cdots+n^3}{2+4+\cdots+2n} = \dfrac{n^2(n+1)^2/4}{n(n+1)} = \dfrac{n(n+1)}{4}$.
2. Last term ka denominator $2+4+\cdots+30$ hai, i.e. $n=15$; so 15 terms hain (30 nahi).
3. $S = \dfrac{1}{4}\sum_{n=1}^{15}(n^2+n) = \dfrac{1}{4}\left(\dfrac{15\cdot 16\cdot 31}{6} + \dfrac{15\cdot 16}{2}\right) = \dfrac{1}{4}(1240+120) = 340.$
4. Check: $\sum n(n+1)/4 = n(n+1)(n+2)/12 = 15\cdot 16\cdot 17/12 = 340.$

Final Answer: (4) 340

Trap: (1) 280 = sirf 14 terms tak joda (last term ka n galat, limit off-by-one); (2) 408 = 16 terms le liye; (3) 680 = denominator ko $1+2+\cdots+n$ samajh kar $T_n = n(n+1)/2$ liya (double value).

Concept to recognise: Each term is (sum of cubes)/(arithmetic sum): reduce the general term with sigma n^3 and the even-number sum to a polynomial in n, then apply sigma n^2, sigma n; the last written term fixes how many terms there are. *(Series sum via sigma formulas)*

Q3. Answer: (1)

Vectors + Circle — Hard

Stem: Let $\vec{a} = 4\hat{i}+\hat{k}$ and $\vec{b} = \hat{j}+\hat{k}$. Let $\vec{c} = c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$, with $c_1c_2c_3 \neq 0$, be a vector such that $\vec{a}\times\vec{c} = 2(\vec{c}\times\vec{b})$. If the circle $x^2+y^2+c_1x+c_2y+c_3=0$ touches the y-axis and meets the x-axis at the points A and B, then the length AB is equal to :

1. $\vec{c}\times\vec{b} = -\vec{b}\times\vec{c}$, so relation banti hai $\vec{a}\times\vec{c}+2(\vec{b}\times\vec{c}) = \vec{0} \Rightarrow (\vec{a}+2\vec{b})\times\vec{c} = \vec{0}$.
2. Isliye $\vec{c} \parallel \vec{a}+2\vec{b} = 4\hat{i}+2\hat{j}+3\hat{k}$, i.e. $\vec{c} = \lambda(4\hat{i}+2\hat{j}+3\hat{k})$, $(c_1,c_2,c_3) = (4\lambda, 2\lambda, 3\lambda)$.
3. Circle $x^2+y^2+4\lambda x+2\lambda y+3\lambda = 0$: $g = 2\lambda$, $f = \lambda$, $c = 3\lambda$. y-axis ko touch kare $\Rightarrow f^2 = c \Rightarrow \lambda^2 = 3\lambda \Rightarrow \lambda = 3$ ($\lambda = 0$ reject, kyunki $c_1c_2c_3 \neq 0$).
4. Circle: $x^2+y^2+12x+6y+9=0$, $g=6$, $c=9$; x-axis par chord $AB = 2\sqrt{g^2-c} = 2\sqrt{36-9} = 6\sqrt{3}$.

Final Answer: (1) $6\sqrt{3}$

Trap: (2) $2\sqrt{3} = \vec{c}\times\vec{b}$ ko $\vec{b}\times\vec{c}$ padh kar $(\vec{a}-2\vec{b})\times\vec{c} = 0$ liya, jisse $\vec{c} = (-4,2,1)$ aur chord $2\sqrt{4-1}$; (3) $6\sqrt{15}$ = chord formula mein $g = c_1 = 12$ rakh diya ($c_1/2$ nahi); (4) $8\sqrt{15}$ = factor 2 drop karke $(\vec{a}+\vec{b})\times\vec{c} = 0$ liya ($\vec{c} = (32,8,16)$).

Concept to recognise: One cross-product relation between an unknown vector c and two known vectors: move both terms to one side (watch the order c x b = -b x c), factor as (combination) x c = 0, so c is parallel to the combination; the circle built from c's components supplies the condition that fixes the scalar. *(Cross-product parallel or coplanarity condition)*

Q4. Answer: (3)

3D Geometry + Vectors — Medium

Stem: Let $\vec{d}$ be a vector such that $\vec{d} \times (\hat{i} + \hat{j} - \hat{k}) = -\hat{i} + \hat{j}$ and $\vec{d} \cdot \hat{k} = -1$. A line L, parallel to $\vec{d}$, meets the lines $L_1 : \frac{x}{2} = \frac{y}{2} = \frac{z}{1}$ and $L_2 : \frac{x}{-2} = \frac{y-3}{1} = \frac{z+3}{2}$ at the points P and Q respectively. Then the length PQ is equal to :

1. $\vec{d} = p\hat{i} + q\hat{j} + r\hat{k}$: $\vec{d} \times (\hat{i} + \hat{j} - \hat{k}) = (-q - r)\hat{i} + (p + r)\hat{j} + (p - q)\hat{k} = -\hat{i} + \hat{j} \Rightarrow p = q$, $q + r = 1$; $\vec{d} \cdot \hat{k} = r = -1 \Rightarrow p = q = 2$, i.e. $\vec{d} = 2\hat{i} + 2\hat{j} - \hat{k}$ (ya $\vec{d} = \frac{\vec{a} \times \vec{b}}{|\vec{a}|^2} + \mu\vec{a}$ se).

2. $P(2s, 2s, s)$ on L_1, $Q(-2t, 3 + t, -3 + 2t)$ on L_2; $\vec{PQ} = (-2t - 2s,\ 3 + t - 2s,\ -3 + 2t - s) \parallel (2, 2, -1)$.

3. Pehle do components barabar: $-2t - 2s = 3 + t - 2s \Rightarrow t = -1$; phir $\frac{2-2s}{2} = \frac{-5-s}{-1} \Rightarrow 1 - s = 5 + s \Rightarrow s = -2$.

4. $P(-4, -4, -2)$, $Q(2, 2, -5)$, $\vec{PQ} = (6, 6, -3) = 3\vec{d}$, so $PQ = 3|\vec{d}| = 9$.

Final Answer: (3) 9

Trap: (1) $4 = L_1, L_2$ ki shortest distance nikaal di, di hui direction ignore karke; (2) 6 = cross product ka order ulta, $(\hat{i} + \hat{j} - \hat{k}) \times \vec{d} = -\hat{i} + \hat{j}$ solve kiya, jisse $\vec{d} = -\hat{k}$ aur $PQ = 6$; (4) 3 = sirf parameter k (ya $|\vec{d}|$) report kiya, $PQ = 3|\vec{d}|$ nahi liya.

Concept to recognise: A line whose direction is defined by a cross-product equation (d x a = b plus a dot condition) 'meets' two given lines: solve the vector equation for d first (componentwise, or d = (a x b)/|a|^2 + mu a), then take general points on both lines, make PQ proportional to d and solve; the length PQ is asked, not the shortest distance. *(Line with given direction meeting two lines)*

Q5. Answer: (2)

Straight Lines — Medium

Stem: Let $L_1 : 3x - 4y + 5 = 0$ and $L_2 : 5x + 12y - 29 = 0$ be two lines. If the bisector of the obtuse angle between L_1 and L_2 meets the coordinate axes at the points P and Q, then the area (in sq. units) of the triangle OPQ, O being the origin, is :

1. Dono lines ko positive constant ke saath likho: $3x - 4y + 5 = 0$ aur $-5x - 12y + 29 = 0$. Yahan $a_1a_2 + b_1b_2 = (3)(-5) + (-4)(-12) = 33 > 0$, so '+' wala bisector obtuse angle ka bisector hai.

2. Obtuse-angle bisector: $\frac{3x - 4y + 5}{5} = \frac{-5x - 12y + 29}{13} \Rightarrow 39x - 52y + 65 = -25x - 60y + 145 \Rightarrow 8x + y - 10 = 0$. (Doosra, $\frac{3x - 4y + 5}{5} = -\frac{-5x - 12y + 29}{13}$, gives $x - 8y + 15 = 0$, the acute-angle bisector.)

3. $8x + y = 10$: $P = (5/4, 0)$, $Q = (0, 10)$.

4. Area $OPQ = \frac{1}{2} \cdot \frac{5}{4} \cdot 10 = \frac{25}{4}$.

Final Answer: (2) $\frac{25}{4}$

Trap: (1) $225/16$ = sign test ignore karke acute-angle bisector $x - 8y + 15 = 0$ le liya (intercepts -15, $15/8$); (4) $9/2$ = normalise kiye bina $L_1 - L_2$ $(x + y = 3)$ ko bisector maan liya; (3) $289/16$ = normalise kiye bina $L_1 + L_2$ $(x + 8y = 17)$ le liya.

Concept to recognise: 'Bisector of the obtuse angle between two given lines': write both lines with positive constants, apply the sign test on a1a2 + b1b2 to pick the right sign in |L1|/sqrt(a1^2+b1^2) = +-|L2|/sqrt(a2^2+b2^2), then use that line (here its intercepts). *(Angle bisector of two lines, internal or obtuse)*

Q6. Answer: (1)

Probability — Medium

Stem: A bag contains 2 white and 4 black balls. A ball is drawn at random from the bag 5 times with replacement. The probability that a white ball is drawn at least 3 times is :

1. Replacement ke saath draws independent hain: $n = 5$, $p = P(\text{white}) = 2/6 = 1/3$, $q = 2/3$.
2. $P(X = r) = {}^5C_r (1/3)^r (2/3)^{5-r}$; 'at least 3' $\Rightarrow r = 3, 4, 5$.
3. $P(X \geq 3) = \dfrac{{}^5C_3 \cdot 4 + {}^5C_4 \cdot 2 + {}^5C_5 \cdot 1}{3^5} = \dfrac{40 + 10 + 1}{243} = \dfrac{51}{243}$.

Final Answer: (1) $\frac{51}{243}$

Trap: (2) 40/243 = 'at least 3' ko 'exactly 3' padh liya; (3) 232/243 = 'at most 3' $(1 - P(X \geq 4))$ nikal diya; (4) 50/243 = $r = 5$ wala term ${}^5C_5(1/3)^5 = 1/243$ jodna bhool gaye (arithmetic slip in the tail sum).

Concept to recognise: 'Put back' / 'with replacement' plus a fixed number of repetitions and 'at least k': binomial distribution, sum the tail terms r = k..n (do not stop at r = n-1). *(Binomial distribution tail probability)*

Q7. Answer: (3)

Complex Numbers — Medium

Stem: Let $\omega \neq 1$ be a cube root of unity. Then $\sum_{k=1}^{30} k \left(\omega^k + \omega^{2k}\right)^3$ is equal to :

1. $\omega^{3k} = 1$ se $\omega^{2k} = \omega^{-k}$; aur $1 + \omega^k + \omega^{2k} = 0$ jab $3 \nmid k$, $= 3$ jab $3 \mid k$.
2. So $\omega^k + \omega^{2k} = 2$ for $3 \mid k$ and $= -1$ for $3 \nmid k$; cube: 8 ya -1 (odd power, sign bacha rehta hai).
3. Multiples of 3 in 1..30: $k = 3, 6, \ldots, 30$ (10 terms), $\sum k = 165$; baaki k ka sum $= 465 - 165 = 300$.
4. Sum $= 8(165) - 300 = 1320 - 300 = 1020$.

Final Answer: (3) 1020

Trap: (1) 750 = $k = 30$ ko multiple of 3 ginna bhool gaye (multiples 3..27 liye); (2) 1620 = $(-1)^3$ ko +1 maan liya (non-multiples ka sign); (4) 1320 = non-multiples wala case bhool gaye (unka contribution 0 maan kar sirf 8×165 liya).

Concept to recognise: A sum over k of a power of (w^k + w^{2k}) with w a cube root of unity: the bracket is periodic with period 3 (values 2 and -1), so split k by residue mod 3; the weight k turns each class into an AP sum. *(Cube roots of unity power sums)*

Q8. Answer: (4)

Matrices & Determinants + Binomial Theorem — Medium

Stem: For a natural number n, let $D(n) = \begin{vmatrix} {}^nC_0 & {}^nC_2 & {}^nC_3 \\ {}^{n+1}C_0 & {}^{n+1}C_2 & {}^{n+1}C_3 \\ {}^{n+2}C_0 & {}^{n+2}C_2 & {}^{n+2}C_3 \end{vmatrix}$, where ${}^mC_r = 0$ for $r > m$. If $\sum_{n=1}^{k} D(n)$ is equal to the coefficient of x in the expansion of $\left(x + \frac{1}{x^2}\right)^{10}$, then k is equal to :

1. Pascal's rule ${}^{m+1}C_r - {}^mC_r = {}^mC_{r-1}$: $R_3 \to R_3 - R_2$, $R_2 \to R_2 - R_1$ se $R_2 = \left(0,\ n,\ \frac{n(n-1)}{2}\right)$, $R_3 = \left(0,\ n+1,\ \frac{n(n+1)}{2}\right)$.
2. C_1 ke along expand: $D(n) = n \cdot \frac{n(n+1)}{2} - (n+1) \cdot \frac{n(n-1)}{2} = \frac{n(n+1)}{2} = {}^{n+1}C_2$.
3. $\sum_{n=1}^{k} {}^{n+1}C_2 = {}^{k+2}C_3$ (hockey-stick; ya $\frac{k(k+1)(k+2)}{6}$ by sigma formulas).
4. $\left(x + \frac{1}{x^2}\right)^{10}$ ka general term $T_{r+1} = {}^{10}C_r\, x^{10-r}\, x^{-2r} = {}^{10}C_r\, x^{10-3r}$; $10 - 3r = 1 \Rightarrow r = 3$, coefficient $= {}^{10}C_3 = 120$.
5. ${}^{k+2}C_3 = 120 = {}^{10}C_3 \Rightarrow k + 2 = 10 \Rightarrow k = 8$.

Final Answer: (4) 8

Trap: (1) 9 = $D(n)$ ko ${}^nC_2$ le liya ya hockey-stick mein ${}^{k+1}C_3$ likh diya (index slip); (2) 10 = ${}^{k+2}C_3 = {}^{10}C_3$ se $k + 2 = 10$ hi answer likh diya; (3) 7 = hockey-stick ka upper index ek zyada, ${}^{k+3}C_3 = 120$. (Binomial step mein $(1/x^2)^r$ ka power $-r$ le lene par $10 - 2r = 1$ ka integer r hi nahi milta.)

Concept to recognise: A 3x3 determinant whose rows are consecutive rows of binomial coefficients, with the sum

of D(n) tied to a coefficient in a binomial expansion: Pascal row operations collapse D(n) to a polynomial in n, the hockey-stick (or sigma) sum gives (k+2)C3, and the general term T_{r+1} of the expansion supplies the number to match. *(Expand determinant with structured entries)*

Q9. Answer: (1)

Permutations & Combinations — Medium

Stem: If all the words (with or without meaning) formed by using all the letters of the word MADRAS are arranged as in a dictionary, then the position of the word MADRAS in this list is :

1. Letters in order: A, A, D, M, R, S (A repeated twice).
2. Words starting with A: baaki A, D, M, R, S distinct $\Rightarrow 5! = 120$. Starting with D: baaki A, A, M, R, S $\Rightarrow 5!/2! = 60$.
3. Now M fixed. Second letter A: isse chhota koi nahi $\Rightarrow 0$. Third letter D (baaki A, D, R, S): A se shuru $\Rightarrow 3! = 6$. Fourth letter R (baaki A, R, S): A $\Rightarrow 2! = 2$. Fifth A (baaki A, S): 0.
4. Rank $= 120 + 60 + 0 + 6 + 2 + 0 + 1 = 189$.

Final Answer: (1) 189

Trap: (2) 249 = D wale block mein repeated A ke liye 2! se divide nahi kiya (120 liya); (3) 129 = A wale block ko bhi 2 se divide kar diya (jabki A fix hone ke baad baaki letters distinct hain); (4) 188 = aakhri +1 (word khud) bhool gaye.

Concept to recognise: 'Arranged as in a dictionary' + 'position of the word' with one letter repeated: alphabetise, count the words before it position by position, dividing by 2! only while the repeated letter is still unused, then add 1. *(Dictionary rank of a word with repeated letter)*

Q10. Answer: (2)

Sequences & Series — Medium

Stem: Let k arithmetic means $A_1, A_2, \ldots, A_k$ be inserted between 2 and 38 such that $A_2 : A_{k-1} = 1 : 4$. Then the sum $A_1 + A_2 + \cdots + A_k$ is equal to :

1. $2, A_1, \ldots, A_k, 38$ ek AP hai with $k+2$ terms, so $d = \dfrac{38-2}{k+1} = \dfrac{36}{k+1}$.
2. $A_2 = 2 + 2d$, $A_{k-1} = 2 + (k-1)d = 38 - 2d$. Condition: $4(2+2d) = 38 - 2d \Rightarrow 10d = 30 \Rightarrow d = 3$.
3. $k + 1 = 36/3 = 12 \Rightarrow k = 11$.
4. Means: $5, 8, \ldots, 35$; sum $= k \cdot \dfrac{2+38}{2} = 11 \times 20 = 220$.

Final Answer: (2) 220

Trap: (1) 240 = $k + 1 = 12$ ko hi k maan liya (12 means); (3) 260 = sum mein 2 aur 38 bhi jod diye (13×20); (4) 200 = $k = 36/3 - 2 = 10$ (off-by-one in the count of terms).

Concept to recognise: 'k arithmetic means inserted between a and b' with a ratio condition on two means: the means form an AP of k+2 terms with d = (b-a)/(k+1); the ratio gives d, then k, then the sum of means = k(a+b)/2. *(AP terms from sum and term conditions)*

Q11. Answer: (4) 3D Geometry + Vectors — Medium

Stem: Let $L : \frac{x-3}{2} = \frac{y-1}{1} = \frac{z}{-2}$ be a line and let $\vec{u}$ be a vector such that $\vec{u} \times (\hat{i}+\hat{j}) = -\hat{i}+\hat{j}+2\hat{k}$ and $\vec{u} \cdot \hat{j} = -1$. If the line through the point $P(\lambda, 5, -5)$, parallel to $\vec{u}$, meets L at the point Q, then PQ^2 is equal to :

1. $\vec{u} = p\hat{i}+q\hat{j}+r\hat{k}$: $\vec{u}\times(\hat{i}+\hat{j}) = -r\hat{i}+r\hat{j}+(p-q)\hat{k} = -\hat{i}+\hat{j}+2\hat{k} \Rightarrow r = 1, p-q = 2$; $\vec{u}\cdot\hat{j} = q = -1 \Rightarrow p = 1$, so $\vec{u} = \hat{i} - \hat{j} + \hat{k}$.
2. $Q(3+2t,\ 1+t,\ -2t)$ on L; $\vec{PQ} = (3+2t-\lambda,\ t-4,\ 5-2t) \parallel (1,-1,1)$: $\frac{3+2t-\lambda}{1} = \frac{t-4}{-1} = \frac{5-2t}{1}$.
3. Last two: $4 - t = 5 - 2t \Rightarrow t = 1$; first: $5 - \lambda = 3 \Rightarrow \lambda = 2$ (isi condition se line L ko milti hai).
4. $P(2, 5, -5)$, $Q(5, 2, -2)$, $\vec{PQ} = (3, -3, 3)$, $PQ^2 = 27$.

Final Answer: (4) 27

Trap: (1) 26 = P se L ki perpendicular distance ka square le liya, di hui direction ignore karke; (2) 43 = λ solve kiye bina $\lambda = 0$ rakh diya (algebra slip); (3) 11 = cross product ka order ulta, $(\hat{i}+\hat{j}) \times \vec{u} = -\hat{i}+\hat{j}+2\hat{k}$ se $\vec{u} = (-3,-1,-1)$ mila, phir $\lambda = 12$, $Q(9,4,-6)$, $PQ^2 = 11$.

Concept to recognise: 'Line through P parallel to u meets L at Q' with u defined by a cross-product equation: solve the vector equation first, then take Q(t) on L and make PQ proportional to u (this also fixes the unknown in P); PQ^2 is asked, not the perpendicular distance. *(Distance measured along a given direction)*

Q12. Answer: (3) Sequences & Series — Medium

Stem: The sum of the infinite series $\frac{1}{6} + \frac{5}{36} + \frac{19}{216} + \frac{65}{1296} + \dots$ is equal to :

1. Numerators $1, 5, 19, 65, \cdots = 3^n - 2^n$ (i.e. $3^{n-1} + 3^{n-2}\cdot 2 + \cdots + 2^{n-1}$, ek GP block), denominators 6^n.
2. $t_n = \dfrac{3^n - 2^n}{6^n} = \left(\dfrac{1}{2}\right)^n - \left(\dfrac{1}{3}\right)^n$.
3. $S = \sum_{n\geq 1}\left(\dfrac{1}{2}\right)^n - \sum_{n\geq 1}\left(\dfrac{1}{3}\right)^n = \dfrac{1/2}{1-1/2} - \dfrac{1/3}{1-1/3} = 1 - \dfrac{1}{2} = \dfrac{1}{2}$.

Final Answer: (3) $\frac{1}{2}$

Trap: (1) 3/4 = doosri GP par $\dfrac{a}{1+r}$ laga diya ($1 - \frac{1/3}{1+1/3}$, formula misuse); (2) 1/3 = dono GPs ke liye ek hi ratio 1/2 use kar li ($1 - \frac{1/3}{1-1/2}$); (4) 1 = poori series ko ek hi GP maan liya with $r = t_2/t_1 = 5/6$ ($\frac{1/6}{1-5/6}$).

Concept to recognise: Numerators growing like a^n - b^n over a power denominator (1, 5, 19, 65 over 6^n): each term is a finite GP block, so split the series into two geometric series and sum each with r/(1-r). *(Series of GP-block terms)*

Q13. Answer: (1) Definite Integration — Medium

Stem: Let $I_n = \int_0^{\pi/4} \tan^n x\, dx$, $n \in \mathbb{N}$. Then $I_3 - I_5$ is equal to :

1. $I_n + I_{n+2} = \int_0^{\pi/4} \tan^n x \sec^2 x\, dx = \left[\dfrac{\tan^{n+1} x}{n+1}\right]_0^{\pi/4} = \dfrac{1}{n+1}$.
2. $I_1 = \int_0^{\pi/4} \tan x\, dx = [\log_e \sec x]_0^{\pi/4} = \log_e \sqrt{2} = \dfrac{1}{2}\log_e 2$.
3. $n = 1$: $I_3 = \dfrac{1}{2} - I_1 = \dfrac{1}{2} - \dfrac{1}{2}\log_e 2$; $n = 3$: $I_5 = \dfrac{1}{4} - I_3 = \dfrac{1}{2}\log_e 2 - \dfrac{1}{4}$.
4. $I_3 - I_5 = \dfrac{1}{2} - \dfrac{1}{2}\log_e 2 - \dfrac{1}{2}\log_e 2 + \dfrac{1}{4} = \dfrac{3}{4} - \log_e 2$ (positive, kyunki $\tan x < 1$ par $\tan^5 x < \tan^3 x$).

Final Answer: (1) $\frac{3}{4} - \log_e 2$

Trap: (2) $\log_e 2 - \frac{1}{2}$ = chain ek step pehle rok di, $I_1 - I_3$ nikal diya; (3) $\frac{1}{2}\log_e 2 - \frac{1}{4}$ = sirf I_5 report kar diya, I_3 se subtract nahi kiya; (4) $1 - \log_e 2 = I_3 - I_5 = 2I_3 - \frac{1}{4}$ mein $-\frac{1}{4}$ chhod diya (sign/term slip).

Concept to recognise: I_n = int_0^{pi/4} tan^n x dx with an odd-index difference asked: pair I_n with I_{n+2} so 1 + tan^2 x = sec^2 x appears (1/(n+1)), start the chain from the exact I_1 = (1/2) ln 2, and keep the alternating

signs straight. *(legacy archetype (writer supplies))*

Q14. Answer: (2) Sets Relations & Functions + Inverse Trigonometry — Medium

Stem: Let $f(x)=\sin^{-1}\left(\frac{2x+[x]}{5}\right)$, where $[x]$ denotes the greatest integer less than or equal to x. If the domain of f is $[\alpha,\beta)$, then $\alpha^2+\beta^2$ is equal to :

1. $\sin^{-1}$ ke liye $-1\le\frac{2x+[x]}{5}\le 1$, i.e. $-5\le 2x+[x]\le 5$.

2. $x\in[n,n+1)$ par $[x]=n$: $-5\le 2x+n\le 5\Rightarrow\frac{-5-n}{2}\le x\le\frac{5-n}{2}$, isko $[n,n+1)$ ke saath intersect karo.

3. $n=-2$: $[-3/2,7/2]\cap[-2,-1)=[-3/2,-1)$; $n=-1$: $[-1,0)$; $n=0$: $[0,1)$; $n=1$: $[-3,2]\cap[1,2)=[1,2)$; $n=2$: $[-7/2,3/2]\cap[2,3)=\emptyset$; $n=-3$: $[-1,4]\cap[-3,-2)=\emptyset$.

4. Domain $=[-3/2,2)$, so $\alpha=-3/2$, $\beta=2$; $\alpha^2+\beta^2=9/4+4=25/4$.

Final Answer: (2) $\frac{25}{4}$

Trap: (4) $50/9=[x]$ ko x maan kar $|3x/5|\le 1$ laga diya ($[-5/3,5/3]$, formula misuse); (1) $5=n=-2$ wala piece chhoot gaya ($\alpha=-1$); (3) $13/4=n=1$ wala piece chhoot gaya ($\beta=1$).

Concept to recognise: sinˆ-1 of a mix of x and [x]: impose -1 <= argument <= 1, split x into unit intervals [n, n+1) where [x] = n, solve each piece and take the union; check the pieces at both ends until they become empty. *(Domain of inverse-trig with greatest integer)*

Q15. Answer: (3) Ellipse — Medium

Stem: Let E be the ellipse whose foci are $(\pm 3\sqrt{3},0)$ and whose minor axis is of length 6. If l is the length of the chord of E whose midpoint is the point $(4,2)$, then l^2 is equal to :

1. $b=3$, $ae=3\sqrt{3}\Rightarrow a^2=b^2+a^2e^2=9+27=36$; $E:\frac{x^2}{36}+\frac{y^2}{9}=1$. $(4,2)$ andar hai: $\frac{16}{36}+\frac{4}{9}=\frac{8}{9}<1$.

2. Midpoint $(4,2)$ wali chord: $T=S_1$: $\frac{4x}{36}+\frac{2y}{9}=\frac{16}{36}+\frac{4}{9}\Rightarrow x+2y=8$ (slope $-\frac{b^2h}{a^2k}=-\frac{1}{2}$).

3. $x=8-2y$ in E: $(8-2y)^2+4y^2=36\Rightarrow 2y^2-8y+7=0\Rightarrow y=2\pm\frac{1}{\sqrt{2}}$, $x=4\mp\sqrt{2}$.

4. $l^2=(2\sqrt{2})^2+(\sqrt{2})^2=8+2=10$.

Final Answer: (3) 10

Trap: (4) 90 = chord ka slope $+\frac{b^2h}{a^2k}=+\frac{1}{2}$ le liya (sign ulta): line $x=2y$ (ek diameter), $l^2=72+18$; (1) 8 = sirf x-coordinates ka difference square kiya, y wala part chhod diya; (2) 40 = half-chord formula se mila l phir se double kar diya ($(2l)^2$).

Concept to recognise: 'Chord of the ellipse whose midpoint is (h, k)': write the chord by T = S1 (or slope -bˆ2 h / aˆ2 k through the midpoint), intersect it with the ellipse and take the distance between the two roots; the ellipse itself may first need aˆ2 from the foci and the minor axis. *(legacy archetype (writer supplies))*

Q16. Answer: (4) Application of Derivatives + Limits — Medium

Stem: Let $f(x)$ be a polynomial of degree 3 having local extrema at $x=2$ and $x=4$, and let $\lim_{x\to 0}\frac{f(x)-3}{x}=24$. Then the local minimum value of f is equal to :

1. Limit finite hone ke liye $f(0)=3$, aur limit $=f'(0)=24$. So $f(x)=ax^3+bx^2+24x+3$.

2. $f'(x)=3ax^2+2bx+24$; $f'(2)=0$, $f'(4)=0$: $12a+4b+24=0$, $48a+8b+24=0\Rightarrow a=1$, $b=-9$.

3. $f(x)=x^3-9x^2+24x+3$, $f'(x)=3(x-2)(x-4)$; $f''(x)=6x-18$: $f''(2)<0$ (maximum), $f''(4)>0$ (minimum).

4. Local minimum value $=f(4)=64-144+96+3=19$.

Final Answer: (4) 19

Trap: (1) 23 = $x = 2$ ko minimum maan liya ($f(2)$, max/min decision galat); (2) 21 = beech ka point $x = 3$ ($f'' = 0$) le liya; (3) 3 = limit se mila $f(0)$ hi report kar diya.

Concept to recognise: A polynomial of stated degree with named extrema plus a limit of (f(x) - c)/x at 0: the limit gives f(0) and f'(0), f' = 0 at the named points gives linear equations for the rest; then classify the two extrema before evaluating. *(Polynomial from extrema and limit conditions)*

Q17. Answer: (2)

Circle + Parabola — Medium

Stem: Let F be the focus of the parabola $x^2 - 4x + 8y - 4 = 0$. Then the minimum distance between F and a point on the circle $x^2 + y^2 - 28x + 12y + 207 = 0$ is :

1. Parabola: $(x-2)^2 = -8(y-1)$, vertex $(2,1)$, $4a = 8 \Rightarrow a = 2$, neeche khulta hai, so focus $F = (2, 1-2) = (2,-1)$.
2. Circle: $(x-14)^2 + (y+6)^2 = 196 + 36 - 207 = 25$; centre $C(14,-6)$, $r = 5$.
3. $CF = \sqrt{12^2 + 5^2} = 13 > r$, so F circle ke bahar hai.
4. Minimum distance $= CF - r = 13 - 5 = 8$ (maximum hota $13 + 5 = 18$).

Final Answer: (2) 8

Trap: (1) 10 = parabola ko upar khulta maan kar focus $(2,3)$ liya ($CF = 15$, $15 - 5$); (3) 18 = minimum ki jagah maximum $CF + r$ le liya; (4) 13 = sirf centre tak ki doori, radius ghataana bhool gaye.

Concept to recognise: 'Minimum/maximum distance between a fixed point and a point on a circle': find the point (here the parabola's focus), the centre and radius, then d - r or d + r; the sign of the parabola's 4a decides which side the focus lies. *(Two-circle position or extreme distance via centre distance)*

Q18. Answer: (1)

Differential Equations + Trigonometry — Medium

Stem: Let $y = y(x)$ be the solution of the differential equation $x\frac{dy}{dx} + \sin 2y = 4x^2\cos^2 y$, $x > 0$, with $y(1) = \frac{3\pi}{4}$. Then $\sin\left(y(\sqrt{2})\right) + \cos\left(y(\sqrt{2})\right)$ is equal to :

1. $\cos^2 y$ se divide ($\frac{\sin 2y}{\cos^2 y} = 2\tan y$): $x\sec^2 y\frac{dy}{dx} + 2\tan y = 4x^2$; $t = \tan y \Rightarrow \frac{dt}{dx} + \frac{2}{x}t = 4x$.
2. I.F. $= x^2$: $\frac{d}{dx}(x^2 t) = 4x^3 \Rightarrow x^2 t = x^4 + C$; $y(1) = \frac{3\pi}{4} \Rightarrow t(1) = -1 \Rightarrow C = -2$, so $\tan y = x^2 - \frac{2}{x^2}$.
3. Hidden condition: $\cos y = 0$ par equation $y' = 0$ deti hai aur $\tan y$ har $x > 0$ par finite hai, so continuous y kabhi $\frac{\pi}{2}$ ya $\frac{3\pi}{2}$ cross nahi karta: $y \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$, i.e. $y = \pi + \tan^{-1}\left(x^2 - \frac{2}{x^2}\right)$. ($x = 2^{1/4}$ par $\tan y = 0$, $y = \pi$: y 2nd quadrant se 3rd mein chala jaata hai.)
4. $x = \sqrt{2}$: $\tan y = 2 - 1 = 1 \Rightarrow y(\sqrt{2}) = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$ (principal value $\frac{\pi}{4}$ nahi).
5. $\sin y + \cos y = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\sqrt{2}$ (check: $(\sin y + \cos y)^2 = 1 + \sin 2y = 2$, third quadrant mein sign negative).

Final Answer: (1) $-\sqrt{2}$

Trap: Hidden condition (sign/direction trap): y continuous hai aur $\cos y = 0$ wali lines cross nahi kar sakta, par $\tan y$ -1 se $+1$ ho jaata hai, so $y(\sqrt{2})$ third quadrant mein hai. (2) $\sqrt{2}$ = ise miss karke principal value $y = \tan^{-1} 1 = \pi/4$ le liya; (3) $0 = y$ ko $y(1)$ wale 2nd quadrant mein hi rakh kar $y = 3\pi/4$ le liya; (4) $-3/\sqrt{5} = \tan(3\pi/4)$ ko $+1$ le liya ($C = 0$, $\tan y(\sqrt{2}) = 2$), branch sahi.

Concept to recognise: sin 2y, cosˆ2 y and y' together: divide by cosˆ2 y and put t = tan y to get a linear DE; then convert tan y back to y on the branch fixed by the initial value (continuity forbids crossing odd multiples of pi/2), so watch whether tan y changes sign on the way. *(Trig substitution reduces nonlinear DE to linear)*

Q19. Answer: (3)

Trigonometry — Medium

Stem: The number of integral values of k for which the equation $8\sin^2 x + 4\cos x = k$ has at least one real solution is :

1. $\sin^2 x = 1 - \cos^2 x$: $k = -8\cos^2 x + 4\cos x + 8 = g(c)$, $c = \cos x \in [-1, 1]$.
2. $g(c) = -8c^2 + 4c + 8$ ka vertex $c = \dfrac{4}{16} = \dfrac{1}{4} \in [-1, 1]$ (maximum): $g(1/4) = -\dfrac{1}{2} + 1 + 8 = \dfrac{17}{2}$.
3. Endpoints: $g(-1) = -8 - 4 + 8 = -4$, $g(1) = -8 + 4 + 8 = 4$. Range of g on $[-1, 1]$: $\left[-4, \dfrac{17}{2}\right]$.
4. Integers k with $-4 \le k \le 8.5$: $-4, -3, \ldots, 8$, i.e. 13 values.

Final Answer: (3) 13

Trap: (1) 9 = vertex ko ignore karke range $[g(-1), g(1)] = [-4, 4]$ le li; (2) 12 = -4 se 8 tak ginne mein $k = 0$ (ya ek end) chhoot gaya; (4) 17 = expression ko $a\sin x + b\cos x$ type maan kar symmetric range $[-17/2, 17/2]$ le li.

Concept to recognise: sin^2 x and cos x together set equal to a parameter: convert to a quadratic in cos x on [-1, 1], check whether the vertex lies inside the interval, take the range, then count the integers in it. *(Range of trig expression and integer parameter count)*

Q20. Answer: (4)

Straight Lines + Sequences & Series — Medium

Stem: Let T_1 be the equilateral triangle with one vertex at $A(1, 1)$ and the opposite side lying on the line $3x+4y+8 = 0$. A triangle T_2 is formed by joining the midpoints of the sides of T_1, a triangle T_3 by joining the midpoints of the sides of T_2, and so on. Then the sum of the areas (in sq. units) of all the triangles $T_1, T_2, T_3, \ldots$ is :

1. Equilateral triangle mein vertex A se opposite side tak ki perpendicular distance hi altitude hai: $h = \dfrac{|3 + 4 + 8|}{\sqrt{9 + 16}} = 3$.
2. Side $a = \dfrac{2h}{\sqrt{3}} = 2\sqrt{3}$, area of $T_1 = \dfrac{\sqrt{3}}{4}a^2 = \dfrac{\sqrt{3}}{4} \cdot 12 = 3\sqrt{3}$ $\left(= \dfrac{h^2}{\sqrt{3}}\right)$.
3. Midpoints jodne se side aadhi hoti hai, so area $\dfrac{1}{4}$ ho jaata hai: areas $3\sqrt{3}, \dfrac{3\sqrt{3}}{4}, \dfrac{3\sqrt{3}}{16}, \ldots$ — infinite GP with $r = \dfrac{1}{4}$.
4. Sum $= \dfrac{3\sqrt{3}}{1 - 1/4} = 4\sqrt{3}$.

Final Answer: (4) $4\sqrt{3}$

Trap: (2) $6\sqrt{3}$ = side aadhi to area bhi aadhi maan kar $r = 1/2$ le liya (algebra slip in the ratio); (1) $3\sqrt{3}$ = sirf T_1 ka area (ya altitude ko hi side maan kar $\frac{\sqrt{3}}{4} \cdot 9 \cdot \frac{4}{3}$); (3) $\sqrt{3}$ = sum T_2 se shuru kiya, T_1 chhod diya.

Concept to recognise: Equilateral triangle fixed by a vertex and the line of the opposite side: the altitude is the perpendicular distance from the vertex to that line; everything else (side, area, centres) follows, and the midpoint-triangle chain is a GP with area ratio 1/4. *(Equilateral triangle from vertex and side line)*

Q21. Answer: 30

Sets Relations & Functions + Logarithms — Medium

Stem: Let $A = \{1, 2, 3\}$ and $B = \{x \in \mathbb{N} : \log_2 x \text{ is a positive integer and } x < 20\}$. Then the number of onto functions $f : B \to A$ such that $f(2) \neq f(4)$ is _____.

1. $\log_2 x \in \{1, 2, 3, 4\}$ with $x < 20 \Rightarrow B = \{2, 4, 8, 16\}$, $|B| = 4$, $|A| = 3$.
2. Total onto functions $B \to A$: $3^4 - {}^3C_1 2^4 + {}^3C_2 1^4 = 81 - 48 + 3 = 36$.
3. Onto functions with $f(2) = f(4)$: common value 3 ways; phir $f(8), f(16)$ ko baaki do values lene hi honge (dono alag), $2! = 2$ ways $\Rightarrow 3 \times 2 = 6$.
4. Required $= 36 - 6 = 30$.

Final Answer: 30

Trap: Counting slip: condition $f(2) \neq f(4)$ ignore karne par 36; $f(2) = f(4)$ wale case ko $3 \times 2^2 = 12$ gin lene par (onto condition bhool kar) 24; inclusion-exclusion ka +3 chhodne par $33 - 6 = 27$. Sab alag integers; sahi 30.

Concept to recognise: 'Number of onto functions' from a 4-element set (built from a one-line log condition) to a 3-element set with one inequality constraint: inclusion-exclusion for the full onto count, then subtract the small complementary case. *(Counting functions between finite sets)*

Q22. Answer: 13440

Binomial Theorem — Medium

Stem: If the sum of the binomial coefficients in the expansion of $\left(x^3 - \frac{2}{x^2}\right)^n$ is 1024, then the term independent of x in this expansion is _____.

1. Binomial coefficients ka sum ${}^nC_0 + {}^nC_1 + \cdots + {}^nC_n = 2^n = 1024 \Rightarrow n = 10$.
2. General term: $T_{r+1} = {}^{10}C_r (x^3)^{10-r} \left(-\frac{2}{x^2}\right)^r = {}^{10}C_r (-2)^r x^{30-5r}$.
3. Independent of x: $30 - 5r = 0 \Rightarrow r = 6$.
4. Term $= {}^{10}C_6 (-2)^6 = 210 \times 64 = 13440$.

Final Answer: 13440

Trap: Formula misuse: $(-2)^r$ ki jagah 2^{n-r} lagane par ${}^{10}C_6 \cdot 2^4 = 3360$; 'sum of coefficients' ko $(1-2)^n$ samajhne par n nikalta hi nahi; exponent $30 - 5r$ ki jagah $30 - 3r = 0$ lene par $r = 10$ aur term $= (-2)^{10} = 1024$. Sahi: 13440.

Concept to recognise: 'Sum of the binomial coefficients is 2^k' hands you n; then the general term of (x^p - c/x^q)^n with the exponent condition p(n-r) - qr = 0 picks r, and the term carries (-c)^r. *(General term coefficient or term independent of x)*

Q23. Answer: 10385

Definite Integration + Sequences & Series — Medium

Stem: The value of $\pi \sum_{r=1}^{30} \int_0^r (x+1) |\cos \pi x| \, dx$ is equal to _____.

1. $|\cos \pi x|$ ka period 1 hai aur $\int_0^1 |\cos \pi x| \, dx = \frac{2}{\pi}$, so $\int_0^r |\cos \pi x| \, dx = \frac{2r}{\pi}$ (r periods).
2. $I_r = \int_0^r x |\cos \pi x| \, dx$: $x \to r - x$ se ($|\cos \pi(r - x)| = |\cos \pi x|$, integer r) $2I_r = r \int_0^r |\cos \pi x| \, dx = \frac{2r^2}{\pi} \Rightarrow I_r = \frac{r^2}{\pi}$.
3. $\int_0^r (x+1) |\cos \pi x| \, dx = \frac{r^2 + 2r}{\pi}$.
4. $\pi \sum_{r=1}^{30} \frac{r^2+2r}{\pi} = \sum_{r=1}^{30} r^2 + 2 \sum_{r=1}^{30} r = \frac{30 \cdot 31 \cdot 61}{6} + 30 \cdot 31 = 9455 + 930 = 10385$.

Final Answer: 10385

Trap: Classic slip: $x|\cos \pi x|$ par plain periodicity laga kar $\int_0^r x|\cos \pi x| = \frac{r}{\pi}$ lena (x factor har period mein shift hota hai) — total $\sum 3r = 1395$; +1 wala part bhoolne par 9455; $\int_0^1 |\cos \pi x| = \frac{1}{\pi}$ lene par $\sum (r^2 + r) = 9920$. 30 terms ke liye $\sum r^2$, $\sum r$ formulas zaroori hain. Sahi: 10385.

Concept to recognise: Integral of (x + c) times a modulus-of-trig with integer period over [0, r], inside a 30-term sum over r: the modulus is periodic (one-period value 2/pi), the x factor needs the x -> r - x symmetry (r^2/pi),

and the outer sum needs the sigma r^2 and sigma r formulas. *(Periodic integrand over many periods)*

Q24. Answer: 27 Area under Curves + Continuity & Differentiability — Medium

Stem: Let $f(x) = \begin{cases} ax^2 + bx, & x \le 1 \\ a^2x - b, & x > 1 \end{cases}$, where $a > 0$, be differentiable at $x = 1$. If A is the area of the region bounded by the curve $y = f(x)$, the x-axis and the lines $x = -1$ and $x = 2$, then $2A$ is equal to _____.

1. Continuity at $x = 1$: $a + b = a^2 - b$; differentiability: $2a + b = a^2$ (left derivative $2ax + b$, right a^2).
2. Second se $b = a^2 - 2a$; pehli mein daalo: $a + a^2 - 2a = a^2 - a^2 + 2a \Rightarrow a^2 - 3a = 0 \Rightarrow a = 3$ ($a = 0$ reject, $a > 0$), $b = 3$.
3. $f(x) = 3x^2 + 3x = 3x(x + 1)$ for $x \le 1$ (negative on $(-1, 0)$, positive on $(0, 1]$) aur $f(x) = 9x - 3 > 0$ for $x > 1$.
4. $A = \left|\int_{-1}^{0}(3x^2 + 3x)dx\right| + \int_{0}^{1}(3x^2 + 3x)dx + \int_{1}^{2}(9x - 3)dx = \frac{1}{2} + \frac{5}{2} + \frac{21}{2} = \frac{27}{2}$.
5. $2A = 27$.

Final Answer: 27

Trap: Counting/ordering slip: $(-1, 0)$ par curve axis ke neeche hai — modulus na lene par $A = 25/2$, $2A = 25$; $[1, 2]$ par bhi pehla piece $3x^2 + 3x$ use karne par $2A = 29$; $a = 0$ wala root lene par $f \equiv 0$ (nonsense). Sab alag integers hain; sahi 27.

Concept to recognise: A two-piece function with unknown parameters 'differentiable at the junction' followed by an area: match values and derivatives at the junction first (reject the degenerate root), then integrate piece by piece, splitting where the curve crosses the x-axis. *(Area after resolving a function condition)*

Q25. Answer: 7 Trigonometry — Medium

Stem: If the sum of all the solutions of the equation $\sin 3x + \sin x = \sin 2x$ in the interval $[0, 2\pi]$ is $k\pi$, then k is equal to _____.

1. $\sin 3x + \sin x = 2\sin 2x \cos x$, so equation: $2\sin 2x \cos x - \sin 2x = 0 \Rightarrow \sin 2x\,(2\cos x - 1) = 0$.
2. $\sin 2x = 0 \Rightarrow x = \frac{n\pi}{2}$: in $[0, 2\pi]$: $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$ (dono endpoints shaamil).
3. $\cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$.
4. Sum $= \left(0 + \frac{1}{2} + 1 + \frac{3}{2} + 2\right)\pi + \left(\frac{1}{3} + \frac{5}{3}\right)\pi = 5\pi + 2\pi = 7\pi \Rightarrow k = 7$.

Final Answer: 7

Trap: Counting slip: $\sin 2x$ se divide karke sirf $\cos x = 1/2$ wale roots rakhne par sum $= 2\pi$, $k = 2$; endpoint $x = 2\pi$ chhodne par $k = 5$; $\sin 2x = 0$ ko $x = n\pi$ padhne par $k = 5$ bhi. Sahi: 7 (total 7 solutions, sum 7π).

Concept to recognise: sin 3x + sin x against sin 2x: sum-to-product gives a factor sin 2x common to both sides — factor, never cancel; list every root of each factor in the closed interval (both endpoints included) and add them. *(Trig equation: count or sum solutions in interval)*

PHYSICS – SOLUTIONS

Q26. Answer: (3)

Kinematics + Work Energy Power — Medium

Stem: A projectile of mass 500 g is projected from the ground with a speed of 20 m s^{-1} at an angle of 60° with the horizontal. The work done by the gravitational force on the projectile from the instant of projection up to the instant when its velocity makes an angle of 45° with the horizontal is (take $g = 10$ m s^{-2}):

1. Components: $u_x = 20\cos 60° = 10$ m s^{-1}, $u_y = 20\sin 60° = 10\sqrt{3}$ m s^{-1}; $v_x = u_x$ poore motion mein constant rehta hai.
2. Velocity 45° par: $|v_y| = v_x = 10$ m s^{-1}, isliye $v^2 = v_x^2 + v_y^2 = 200$ aur $K = \frac{1}{2}(0.5)(200) = 50$ J.
3. $K_0 = \frac{1}{2}(0.5)(400) = 100$ J; work-energy theorem: $W_g = \Delta K = 50 - 100 = -50$ J (check: us instant height $y = (u_y^2 - v_y^2)/2g = (300 - 100)/20 = 10$ m, $-mgy = -50$ J).

Final Answer: option (3) -50 J.

Trap: (1) = sign ulta (gravity ko positive work); (2) = 45° wale instant par sirf $v_x = 10$ le liya ($K = 25$ J, jo top wali KE hai) ya instant ko top samajh liya; (4) = interval 45° instant se top tak le liya ($50 - 25$); mass 500 g ko kg mein na badalne par -5×10^4 J aata hai, jo option nahi hai.

Concept to recognise: Launch speed and angle given with 'velocity makes 45° with the horizontal' and 'work done by gravity': the projectile-components archetype; the horizontal component is conserved, so at 45° both components equal $u\cos\theta$, and $W_g = \Delta K$. *(Projectile components, KE and height at a point)*

Q27. Answer: (1)

Thermodynamics — Medium

Stem: An ideal gas is taken from state A to state B along the path ACB shown in the P-V diagram. It absorbs 96 J of heat and does 36 J of work. Along the path ADB, the work done by the gas is 16 J. When the gas is brought back from B to A along the curved path, 24 J of work is done on the gas. The heat absorbed by the gas along ADB and the heat released by the gas along the curved path are, respectively:

1. Path ACB: $\Delta U_{AB} = Q - W = 96 - 36 = 60$ J. Internal energy state function hai, har path par same.
2. Path ADB: $Q = \Delta U + W = 60 + 16 = 76$ J absorbed.
3. Curved return $B \to A$: $\Delta U = -60$ J; work ON the gas 24 J ka matlab work BY the gas $W = -24$ J.
4. $Q = \Delta U + W = -60 - 24 = -84$ J, yaani 84 J heat released.

Final Answer: option (1) 76 J and 84 J.

Trap: (2) = return path mein 'work on gas' ko +24 le liya ($-60 + 24 = -36$); (3) = ADB mein W ka sign ulta ($60 - 16 = 44$); (4) = dono sign slips ek saath.

Concept to recognise: Two named paths between the same two states on a P-V diagram plus a return path phrased as 'work done ON the gas': the first-law bookkeeping archetype where ΔU is path-independent and every sign is set by the convention. *(legacy archetype (writer supplies))*

Q28. Answer: (2)

Oscillations + Laws of Motion — Medium

Stem: A block of mass 8 kg rests on a smooth horizontal surface and is attached to a horizontal spring of spring constant 1000 N m^{-1}. A second block of mass 2 kg is placed on top of the first block. The coefficient of friction between the two blocks is 0.5. The maximum amplitude of simple harmonic motion of the system for which the upper block does not slip on the lower block is (take $g = 10$ m s^{-2}):

1. Dono blocks saath oscillate karte hain: $\omega^2 = k/(M+m) = 1000/10 = 100$ s^{-2}.
2. Upper block par horizontal force sirf friction hai: $f = ma \le \mu mg$, isliye $a_{max} = \omega^2 A \le \mu g$.
3. $A_{max} = \mu g/\omega^2 = \mu(M+m)g/k = (0.5)(10)(10)/1000 = 0.05$ m $= 5$ cm.

Final Answer: option (2) 5 cm.

Trap: (1) $= \omega^2$ mein sirf $M = 8$ kg liya ($\mu Mg/k = 4$ cm); (3) = sirf upper mass m liya ($\mu mg/k = 1$ cm); (4) $= \mu$ chhod kar vertical-platform wali condition $a_{max} = g$ laga di (10 cm).

Concept to recognise: Two stacked blocks, spring on the lower one, a friction coefficient between them and 'does not slip': the stacked-block SHM archetype, where the slip condition is $\omega^2 A \le \mu g$ with ω of the combined mass. *(legacy archetype (writer supplies))*

Q29. Answer: (4)

Units & Measurements — Easy

Stem: The energy E of a particle at position x moving with speed v is given by $E = Ax^2 + \dfrac{B}{x} + Cv$, where A, B and C are constants. The dimensional formula of $\dfrac{BC}{A}$ is:

1. Har term ka dimension E ke barabar hona chahiye: $[E] = [\mathrm{M\,L^2\,T^{-2}}]$.
2. $[A] = [E]/[x^2] = [\mathrm{M\,T^{-2}}]$; $[B] = [E][x] = [\mathrm{M\,L^3\,T^{-2}}]$; $[C] = [E]/[v] = [\mathrm{M\,L\,T^{-1}}]$.
3. $\left[\dfrac{BC}{A}\right] = \dfrac{[\mathrm{M\,L^3\,T^{-2}}][\mathrm{M\,L\,T^{-1}}]}{[\mathrm{M\,T^{-2}}]} = [\mathrm{M\,L^4\,T^{-1}}]$.

Final Answer: option (4) $[\mathrm{M\,L^4\,T^{-1}}]$.

Trap: (1) $= [C]$ mein time exponent ka slip ($\mathrm{T^{-2}/T^{-1}}$ ko $\mathrm{T^{-3}}$ likh diya); (2) = ratio ulta le liya (A/BC); (3) $= A$ se divide karna bhool gaye (sirf $[BC]$).

Concept to recognise: An energy equation with added terms and unknown constants, asked a product or ratio of the constants: the homogeneity archetype, where each term carries $[E]$ and each constant is $[E]$ divided by its own variable. *(Dimensional homogeneity of equation constants)*

Q30. Answer: (1)

Gravitation — Easy

Stem: Two planets A and B are spheres of uniform density. The radius of B is three times the radius of A, and the mean density of B is four times that of A. If a satellite revolving close to the surface of A has a period T, then the period of a satellite revolving close to the surface of B is:

1. Near-surface orbit: $T^2 = \dfrac{4\pi^2 R^3}{GM}$ aur $M = \frac{4}{3}\pi R^3 \rho$.
2. R^3 cancel ho jaata hai: $T = \sqrt{\dfrac{3\pi}{G\rho}}$; period sirf density par depend karta hai, radius par nahi.
3. $\dfrac{T_B}{T_A} = \sqrt{\dfrac{\rho_A}{\rho_B}} = \sqrt{\dfrac{1}{4}} = \dfrac{1}{2}$, so $T_B = T/2$.

Final Answer: option (1) $T/2$.

Trap: (2) $= T \propto \sqrt{\rho}$ ulta laga diya ($2T$); (3) = density ka data chhod kar $T^2 \propto R^3/M$ mein $M \propto R^3$ se T same maan liya; (4) $= M \propto \rho R^2$ likh kar $T^2 \propto 27/36$ se $\frac{\sqrt{3}}{2}T$ (formula misuse).

Concept to recognise: A satellite 'close to the surface' of two planets with both the radius ratio and the density ratio given: the near-surface period archetype in which R cancels and only ρ survives. *(Satellite orbit energy and*

period)

Q31. Answer: (3)

Current Electricity — Medium

Stem: In the circuit shown in the figure, the cells have negligible internal resistance. The heat produced in the $3\,\Omega$ resistor in 10 s is:

1. $V_{AB} = V$ lo. Node A par KCL: $\dfrac{12-V}{2} + \dfrac{4-V}{2} = \dfrac{V}{3}$.
2. $3(12-V) + 3(4-V) = 2V \Rightarrow 48 = 8V \Rightarrow V = 6$ V.
3. $3\,\Omega$ mein current $I = 6/3 = 2$ A (12 V cell 3 A deta hai, jisme se 1 A wapas 4 V cell ko charge karta hai).
4. $H = I^2Rt = (2)^2(3)(10) = 120$ J.

Final Answer: option (3) 120 J.

Trap: (1) = 12 V cell wala current 3 A hi $3\,\Omega$ mein maan liya ($3^2 \cdot 3 \cdot 10$); (2) = $3\,\Omega$ ke across poore 12 V maan kar V^2t/R; (4) = I^2 ki jagah I ($IRt = 60$ J), arithmetic slip.

Concept to recognise: Two cells with series resistors on the outer branches and a bare resistor on the middle rung: the two-loop Kirchhoff archetype with one unknown node voltage; the asked heat then needs I^2Rt. *(Kirchhoff multi-loop network)*

Q32. Answer: (2)

Magnetic Effects of Current + Electrostatics — Medium

Stem: Two large parallel plates P and Q, placed 1 cm apart, are kept at potentials +150 V and −50 V respectively. A uniform magnetic field of 0.05 T exists in the region between the plates, parallel to the plates. A particle of mass 1.6×10^{-27} kg and charge $+1.6 \times 10^{-19}$ C is released from rest close to plate P. Neglecting gravity, the speed with which the particle strikes plate Q is:

1. Magnetic force $q\vec{v} \times \vec{B}$ hamesha $\vec{v}$ ke perpendicular hai: path mud jaata hai, par $\vec{B}$ koi work nahi karta. Speed sirf electric field se badalti hai.
2. Electrostatics step: positive charge higher potential (P) se lower potential (Q) ki taraf jaata hai; $\Delta V = V_P - V_Q = 150 - (-50) = 200$ V aur field ka work $W = q\Delta V$.
3. $\frac{1}{2}mv^2 = q\Delta V \Rightarrow v = \sqrt{2(q/m)\Delta V} = \sqrt{2 \times 10^8 \times 200} = 2 \times 10^5$ m s^{-1}.

Final Answer: option (2) 2×10^5 m s^{-1}.

Trap: (1) = $\Delta V = 150 - 50 = 100$ V le liya (−50 ka sign chhoot gaya), sign slip; $\frac{1}{2}$ bhoolne par bhi yahi aata hai; (3) = sirf $V_P = 150$ V ko potential difference maan liya; (4) = crossed fields dekh kar velocity-selector wala $v = E/B = (\Delta V/d)/B = 4 \times 10^5$ m s^{-1} laga diya.

Concept to recognise: Plates at stated potentials with a magnetic field also present and the speed on striking the other plate asked: the zero-work-of-the-magnetic-force archetype; only $q\Delta V$ between the plates changes the kinetic energy, so B and d never enter. *(Magnetic force zero-work shortcut)*

Q33. Answer: (4)

Fluid Mechanics + Kinematics — Medium

Stem: A cylindrical vessel standing on a block of height 80 cm contains water filled up to a height of 125 cm. A small hole is made in the side wall of the vessel at a height of 45 cm above the base of the vessel. The horizontal distance from the hole at which the jet of water coming out of the hole strikes the ground is equal to : (take $g = 10$ m s^{-2})

1. Hole surface se $h = 125 - 45 = 80$ cm $= 0.8$ m neeche hai: Torricelli $v = \sqrt{2gh} = \sqrt{2(10)(0.8)} = 4$ m s^{-1}.
2. Hole ground se $y = 45 + 80 = 125$ cm $= 1.25$ m upar hai (block bhi count hota hai): $t = \sqrt{2y/g} = \sqrt{0.25} = 0.5$ s.
3. $R = vt = 4 \times 0.5 = 2$ m (ya $R = 2\sqrt{hy} = 2\sqrt{0.8 \times 1.25} = 2$ m).

Final Answer: option (4) 2 m.

Trap: (1) = block bhool kar fall height sirf 45 cm li ($2\sqrt{0.8 \times 0.45} = 1.2$ m); (2) = depth ki jagah poora water level 1.25 m le liya ($2\sqrt{1.25 \times 1.25} = 2.5$ m); (3) = fall height sirf block ki height 0.8 m li, hole ki vessel-base se 45 cm wali height bhool gaye ($2\sqrt{0.8 \times 0.8} = 1.6$ m).

Concept to recognise: A vessel on a block with the water level and the hole height both in cm and 'the jet strikes the ground': the Torricelli-plus-projectile archetype; the fall height must include the block and the depth is level minus hole height. *(Torricelli efflux with projectile range)*

Q34. Answer: (1)

Centre of Mass & Collisions — Medium

Stem: Particles of masses 3 kg, 1 kg, 1 kg, 1 kg and 1 kg are placed at the vertices A, B, D, E and F respectively of a regular hexagon $ABCDEF$ of side a, and there is no particle at the vertex C, as shown in the figure. The distance of the centre of mass of the system from the centre O of the hexagon is:

1. Regular hexagon mein centre se har vertex ki doori side ke barabar hoti hai ($OA = OC = a$). Har vertex par 1 kg hota to symmetry se COM O par hota.

2. Diya gaya system = woh symmetric system + A par extra 2 kg + C par -1 kg (missing particle). O se: $\sum m\vec{r} = 2a\,\hat{r}_A - a\,\hat{r}_C$.

3. $\angle AOC = 120°$: $|\sum m\vec{r}|^2 = 4a^2 + a^2 - 4a^2\cos 120° = 7a^2$, so $|\sum m\vec{r}| = \sqrt{7}\,a$.

4. Total mass $= 3 + 1 + 1 + 1 + 1 = 7$ kg: $r_{cm} = \dfrac{\sqrt{7}\,a}{7}$ $(= a/\sqrt{7})$.

Final Answer: option (1) $\dfrac{\sqrt{7}\,a}{7}$.

Trap: (2) = $\angle AOC$ ko 60° maan liya ($5 - 2 = 3$, so $\sqrt{3}a/7$); (3) = C ko A ka opposite vertex maan kar $2a + a = 3a$ seedha jod diya; (4) = excess/missing-mass method mein denominator mein purana total 6 kg hi rakh diya ($\sqrt{7}a/6$).

Concept to recognise: Particles at the vertices of a regular hexagon with one heavier and one missing: the discrete-COM archetype in a symmetric disguise; start from the symmetric hexagon (COM at the centre) and add the excess and the missing mass as vectors from the centre. *(Centre of mass of discrete masses)*

Q35. Answer: (2)

Rotational Motion + Kinematics — Medium

Stem: Two blocks of masses 3 kg and 1 kg are connected by a light inextensible string passing over a pulley which is a uniform disc of mass 2 kg mounted on a frictionless horizontal axle. The string does not slip on the pulley. At $t = 0$, the 3 kg block is moving vertically upwards with a speed of 10 m s^{-1}. The distance travelled by the 3 kg block in the first 4 s is (take $g = 10$ m s^{-2}):

1. Disc pulley: $I/R^2 = M/2 = 1$ kg. $a = \dfrac{(m_1 - m_2)g}{m_1 + m_2 + I/R^2} = \dfrac{(3-1)(10)}{3+1+1} = 4$ m s^{-2}; 3 kg block ke liye yeh hamesha neeche ki taraf hai, chahe block upar jaa raha ho ya neeche.

2. Kinematics (reversal): velocity acceleration ke opposite hai, block $t_1 = u/a = 10/4 = 2.5$ s par rukta hai; tab tak upar distance $u^2/2a = 100/8 = 12.5$ m.

3. Baaki $4 - 2.5 = 1.5$ s mein rest se neeche: $\frac{1}{2}(4)(1.5)^2 = 4.5$ m.

4. Distance $= 12.5 + 4.5 = 17$ m (displacement sirf $10(4) - \frac{1}{2}(4)(4)^2 = 8$ m upar).

Final Answer: option (2) 17 m.

Trap: (1) = $s = ut - \frac{1}{2}at^2$ wala displacement 8 m hi distance maan liya (reversal ignore kiya); (3) = pulley ka I/R^2 term chhod diya ($a = 5$, $t_1 = 2$ s, $10 + 10 = 20$ m); (4) = sirf turning point tak ka distance $u^2/2a = 12.5$ m liya, block ko wahin ruka maan liya.

Concept to recognise: Two hanging masses over a pulley given a mass and a shape, an initial velocity opposite to the net pull and 'distance travelled': the massive-pulley Atwood archetype with a kinematics reversal; find a with I/R^2, then split the motion at $t = u/a$. *(Atwood machine with massive pulley)*

Q36. Answer: (3)

Thermal Properties — Medium

Stem: A composite rod is made by joining end to end a brass rod of length 60 cm and a steel rod of length 40 cm at 20 °C. The coefficients of linear expansion of brass and steel are 2.0×10^{-5} °C^{-1} and 1.2×10^{-5} °C^{-1} respectively. The length of the composite rod at 120 °C is:

1. $\Delta T = 100\,^\circ$C; har rod ka expansion alag se: $\Delta L = L\alpha\Delta T$.
2. Brass: $60 \times 2.0 \times 10^{-5} \times 100 = 0.120$ cm; steel: $40 \times 1.2 \times 10^{-5} \times 100 = 0.048$ cm.
3. Total: $L = 100 + 0.120 + 0.048 = 100.168$ cm.

Final Answer: option (3) 100.168 **cm.**

Trap: (1) = lengths swap ho gayi (40 cm brass, 60 cm steel: $0.080 + 0.072 = 0.152$); (2) = poore rod par brass ka α laga diya (0.200 cm); (4) = decimal slip, ΔL ko 10 guna (1.68 cm), arithmetic.

Concept to recognise: Two rods of different materials joined end to end and heated through a stated temperature change: the composite-expansion archetype; the expansions add, each with its own $L\alpha$. *(Linear expansion composite or successive)*

Q37. Answer: (4)

Units & Measurements — Medium

Stem: If force F, velocity v and time T are taken as the fundamental quantities, then match List-I with List-II. List-I: (A) Mass; (B) Energy; (C) Density; (D) Linear momentum. List-II: (I) $[F\,v\,T]$; (II) $[F\,T]$; (III) $[F\,v^{-1}\,T]$; (IV) $[F\,v^{-4}\,T^{-2}]$. Choose the correct answer from the options given below:

1. $[F] = \mathrm{MLT}^{-2}$, $[v] = \mathrm{LT}^{-1}$; isliye $\mathrm{L} = vT$ aur $\mathrm{M} = F/\text{acceleration} = FT/v$: Mass $= [F\,v^{-1}\,T]$, (III).
2. Energy $= F\times$ length $= F \cdot vT$: $[F\,v\,T]$, (I).
3. Density $= \mathrm{M/L}^3 = (Fv^{-1}T)/(vT)^3 = [F\,v^{-4}\,T^{-2}]$, (IV).
4. Momentum $= F\times$ time $= [F\,T]$, (II).

Final Answer: option (4) (A)-(III), (B)-(I), (C)-(IV), (D)-(II).

Trap: (1) = mass ke liye v ki power ka sign ulat gaya (FT/v ki jagah FvT), phir energy ko bacha hua $[Fv^{-1}T]$ de diya; (2) = impulse Ft ko energy aur $F \cdot vt$ (= force × distance) ko momentum maan liya, yaani energy-momentum entries interchange; (3) = momentum $p = mv = (Fv^{-1}T)(v) = FT$ compute kiye bina density aur momentum ki bachi hui entries aapas mein badal di (C-D interchange).

Concept to recognise: 'If force, velocity and time are taken as fundamental' with a match-list of mechanical quantities: the re-basing archetype; write $\mathrm{L} = vT$ and $\mathrm{M} = FT/v$ once, then substitute. *(Dimensional analysis exponent solving)*

Q38. Answer: (2)

Capacitance — Medium

Stem: A parallel plate capacitor with air between its plates is charged by a battery and then disconnected from it. A dielectric slab of dielectric constant $K = 3$ and thickness equal to half the plate separation is now inserted so that it covers the entire plate area, as shown in the figure. The ratio of the energy stored in the capacitor after the insertion of the slab to that before is:

1. Slab ke saath gap do series pieces mein bat jaata hai, air $d/2$ aur dielectric $d/2$: $C = \dfrac{\varepsilon_0 A}{d - t + t/K} = \dfrac{\varepsilon_0 A}{d/2 + d/6} = \dfrac{3}{2}C_0$ (general: $\dfrac{2K}{K+1}C_0$).
2. Battery disconnected hai: charge Q fixed, isliye $U = \dfrac{Q^2}{2C} \propto \dfrac{1}{C}$.
3. $\dfrac{U_{after}}{U_{before}} = \dfrac{C_0}{C} = \dfrac{2}{3}$.

Final Answer: option (2) $\dfrac{2}{3}$.

Trap: (1) = battery connected maan kar $U = \frac{1}{2}CV^2 \propto C$ se 3/2; (3) = slab ko area-wise (parallel) split maan liya, $C = \frac{K+1}{2}C_0 = 2C_0 \Rightarrow 1/2$; (4) = poora gap K se bhara maan liya ($C = 3C_0 \Rightarrow 1/3$).

Concept to recognise: 'Charged and then disconnected' plus a slab of half the gap thickness over the full area: the partial-slab (series) capacitor archetype with the isolated-capacitor energy twist; Q stays, C rises, U falls. *(Dielectric-filled capacitor: partial slab or split regions)*

Q39. Answer: (1)

Thermal Properties — Easy

Stem: A large piece of ice at 0 °C is put into 300 g of water at 40 °C contained in a vessel of negligible heat capacity. When thermal equilibrium is reached, some ice is still left unmelted. The mass of ice that has melted is (latent heat of fusion of ice = 80 cal g^{-1}, specific heat of water = 1 cal $g^{-1}\,°C^{-1}$):

1. Kuch barf bachi hai, isliye equilibrium temperature 0 °C hai.
2. Paani ne di hui heat: $300 \times 1 \times (40 - 0) = 12000$ cal.
3. Yeh poori heat sirf melting mein jaati hai: $m \times 80 = 12000 \Rightarrow m = 150$ g.

Final Answer: option (1) 150 g.

Trap: (2) = ratio ulta le liya ($m = 300 \times 80/40 = 600$ g), algebra slip; (3) = aadat se pighli barf ka warming term bhi jod diya, $m(80 + 40) = 12000$ (100 g), jabki final temperature 0 °C hai; (4) = melted ice ki jagah equilibrium par kul paani $300 + 150 = 450$ g likh diya.

Concept to recognise: Ice dropped into warm water with 'some ice is still left unmelted': the heat-balance archetype in which the leftover ice pins the final temperature at 0 °C, so all the heat from the water goes into melting. *(Calorimetry: staged heating and heat balance)*

Q40. Answer: (3)

Nuclei — Medium

Stem: A nucleus of mass number 224, initially at rest, decays by emitting an α-particle. The Q-value of the decay is 6.16 MeV. Treating the nuclear masses as proportional to the mass numbers, the kinetic energies of the α-particle and of the daughter nucleus are, respectively:

1. Parent at rest: momentum conservation $p_\alpha = p_d$, aur $K = p^2/2m \Rightarrow K \propto 1/m$; lighter α ko zyada energy milti hai.

2. $K_\alpha/K_d = m_d/m_\alpha = 220/4$ aur $K_\alpha + K_d = Q = 6.16$ MeV, isliye $K_d = Q\dfrac{4}{A}$ (denominator mein $m_\alpha + m_d = A$, sirf $A-4$ nahi).

3. $K_\alpha = Q\dfrac{A-4}{A} = 6.16 \times \dfrac{220}{224} = 6.05$ MeV; $K_d = 6.16 \times \dfrac{4}{224} = 0.11$ MeV.

Final Answer: option (3) 6.050 **MeV and** 0.110 **MeV.**

Trap: (1) = 'heavier nucleus ko zyada energy' wali ulti soch (energies swap); (2) = Q ko barabar baant diya; (4) = daughter ka share $4/(A-4) = 4/220$ le liya ($4/A$ ki jagah): $K_d = 0.112$, $K_\alpha = 6.048$ MeV; (3) aur (4) dono alpha-majority hain, sirf exact calculation alag karti hai.

Concept to recognise: Parent at rest with the Q-value and mass number given and both daughters' energies asked: the alpha-recoil archetype; equal momenta, so the energies split inversely as the masses. *(Alpha decay energy sharing)*

Q41. Answer: (4)

Semiconductors — Medium

Stem: In the circuit shown in the figure, D_1 and D_2 are ideal diodes and the battery has negligible internal resistance. The current drawn from the battery is:

1. Top rail battery ke + se juda hai (higher potential). D_1 ka p-side top par hai: forward biased, short. D_2 ka p-side bottom par hai: reverse biased, open.

2. Circuit = $4\,\Omega$ series $6\,\Omega$ ($12\,\Omega$ wali branch cut off): $R = 10\,\Omega$.

3. $I = 12/10 = 1.2$ A.

Final Answer: option (4) 1.2 **A.**

Trap: (1) = dono diodes conducting maan liye ($6 \parallel 12 = 4$, $I = 12/8 = 1.5$ A); (2) = D_1 ko reverse aur D_2 ko forward samjha ($12/16 = 0.75$ A), direction slip; (3) = series $4\,\Omega$ bhool gaye ($12/6 = 2$ A).

Concept to recognise: Two diodes drawn in opposite orientations on parallel branches of a DC circuit with 'ideal diodes': the conduction-state archetype; decide the bias of each from the battery polarity, then plain Ohm's law. *(Diode conduction state in DC circuit)*

Q42. Answer: (1)

Fluid Mechanics — Easy

Stem: Two soap bubbles of radii 4 cm and 6 cm, blown from the same soap solution, come in contact and share a common surface. The radius of curvature of the common surface is:

1. Bubble ke andar excess pressure $4T/r$: chhote bubble mein pressure zyada hota hai.

2. Common wall ke across pressure difference $= \dfrac{4T}{r_1} - \dfrac{4T}{r_2} = \dfrac{4T}{r_c}$.

3. $r_c = \dfrac{r_1 r_2}{r_2 - r_1} = \dfrac{4 \times 6}{6-4} = 12$ cm (wall bade bubble ki taraf bulge karti hai).

Final Answer: option (1) 12 **cm.**

Trap: (2) = denominator mein $r_1 + r_2$ le liya ($24/10 = 2.4$ cm); (3) = sirf $r_2 - r_1 = 2$ cm likh diya; (4) = bubbles par $4T/r$ aur common wall par drop wala $2T/r$ mix kar ke $2r_1r_2/(r_2 - r_1) = 24$ cm.

Concept to recognise: Two bubbles 'in contact' with a 'common surface' and its radius of curvature asked: the shared-wall archetype; the wall carries the pressure difference, so the reciprocals of the radii subtract. *(Two bubbles in contact common surface)*

Q43. Answer: (2) Electromagnetic Induction + Current Electricity — Medium

Stem: A conducting rod of length 0.5 m and resistance 1 Ω slides with a constant velocity of 12 m s^{-1} on two long, parallel, smooth conducting rails of negligible resistance, in a uniform magnetic field of 0.5 T perpendicular to the plane of the rails. The rails are connected at one end through a 3 Ω resistor and at the other end through a 6 Ω resistor. The potential difference between the ends of the rod is:

1. Motional emf: $\varepsilon = Blv = 0.5 \times 0.5 \times 12 = 3$ V; rod ek cell ki tarah hai jiska internal resistance $r = 1\,\Omega$ hai.
2. Current Electricity step: 3 Ω aur 6 Ω rod ke dono taraf alag loops banate hain aur dono rails ke beech jude hain, isliye rod ke terminals ke across parallel: $R_p = \dfrac{3 \times 6}{3+6} = 2\,\Omega$.
3. $I = \dfrac{\varepsilon}{R_p + r} = \dfrac{3}{2+1} = 1$ A.
4. Terminal p.d. $V = \varepsilon - Ir = 3 - 1 = 2$ V $(= IR_p)$.

Final Answer: option (2) 2 V.

Trap: (1) = rod ka resistance ignore karke p.d. ko emf 3 V hi maan liya; (3) = 3 Ω aur 6 Ω ko rod ke saath ek series loop maan liya ($I = 3/10 = 0.3$ A, $V = 0.3 \times 9 = 2.7$ V); (4) = rod ke andar ka drop $Ir = 1$ V hi p.d. maan liya.

Concept to recognise: A rod with its own resistance sliding on rails closed by a resistor at each end, with the potential difference across the rod asked: the motional-emf archetype with the rod acting as a cell; the end resistors are in parallel across it, so $V = \varepsilon - Ir$. *(Motional emf of translating conductor)*

Q44. Answer: (4) Oscillations + Fluid Mechanics — Medium

Stem: A uniform wooden cylinder of length 2.5 m and density 640 kg m^{-3} floats vertically in water of density 1000 kg m^{-3}. It is pushed down slightly and released. The time period of the resulting vertical oscillations is (take $g = 10$ m s^{-2}):

1. Floating condition (fluid step): $\rho_b ALg = \rho_w Alg \Rightarrow l = \dfrac{640}{1000} \times 2.5 = 1.6$ m submerged.
2. x aur dabane par extra buoyancy $\rho_w Agx$ restoring force hai: $k = \rho_w Ag$, aur $M = \rho_b AL = \rho_w Al$.
3. $T = 2\pi\sqrt{\dfrac{M}{\rho_w Ag}} = 2\pi\sqrt{\dfrac{l}{g}} = 2\pi\sqrt{\dfrac{1.6}{10}} = 2\pi(0.4) = 0.8\pi$ s.

Final Answer: option (4) 0.8π s.

Trap: (1) = submerged length ki jagah poori length $L = 2.5$ m le li ($2\pi\sqrt{0.25} = \pi$); (2) = 2π ki jagah π (0.4π); (3) = paani ke bahar wala hissa 0.9 m le liya ($2\pi\sqrt{0.09} = 0.6\pi$).

Concept to recognise: A floating cylinder 'pushed down and released' with densities and the length given: the buoyancy-restoring SHM archetype; the period reduces to $2\pi\sqrt{l_{sub}/g}$ with l_{sub} from the floating condition. *(Floating body SHM period)*

Q45. Answer: (3) Electromagnetic Waves — Medium

Stem: The electric field of a plane electromagnetic wave travelling along the positive x-direction is given by $\vec{E} = 90\sin(kx - \omega t)\,(-\hat{j})$ V m^{-1}. Taking $c = 3 \times 10^8$ m s^{-1}, the magnetic field of the wave is:

1. $B_0 = E_0/c = 90/(3 \times 10^8) = 3 \times 10^{-7}$ T; $\vec{E}$ aur $\vec{B}$ same phase mein, same argument $(kx - \omega t)$.
2. Direction: $\vec{E} \times \vec{B}$ propagation $(+\hat{i})$ ki taraf hona chahiye. $\hat{E} = -\hat{j}$ ke liye $(-\hat{j}) \times (-\hat{k}) = \hat{j} \times \hat{k} = \hat{i}$, isliye $\hat{B} = -\hat{k}$.
3. $\vec{B} = 3 \times 10^{-7}\sin(kx - \omega t)\,(-\hat{k})$ T.

Final Answer: option (3).

Trap: (1) = cross product ka order ulta ($\vec{B} \times \vec{E}$ ko propagation maan liya, sign $+\hat{k}$), direction slip; (2) = E_0 ko c se multiply kar diya (2.7×10^{10}), formula misuse; (4) = $\vec{B}$ ko propagation direction $\hat{i}$ ke along maan liya.

Concept to recognise: One field of a plane EM wave given as a sinusoid with a unit vector and the propagation axis stated: the companion-field archetype; divide by c, keep the phase, fix the direction from $\vec{E} \times \vec{B}$ along propagation. *(Plane EM wave E–B vector relation)*

Q46. Answer: 5

Electromagnetic Induction + Alternating Current — Medium

Stem: A circular coil of 100 turns and area 25 cm^2 has a resistance of 10 Ω. It is placed with its plane perpendicular to a magnetic field which varies with time as $B = 0.2\sin(200t)$ T, where t is in seconds. The average power dissipated in the coil is _____ W.

1. Flux $\Phi = NBA$; $\varepsilon = -NA\dfrac{dB}{dt} = -(100)(25 \times 10^{-4})(0.2)(200)\cos(200t) = -10\cos(200t)$ V, yaani $\varepsilon_0 = 10$ V.
2. Induced current sinusoidal hai: $I_0 = \varepsilon_0/R = 1$ A; AC step: $I_{rms} = I_0/\sqrt{2}$.
3. $P_{avg} = I_{rms}^2 R = \dfrac{1}{2}(1)^2(10) = 5$ W.

Final Answer: 5

Trap: $P = I_0^2 R = 10$ W (peak ko rms maan liya, factor $\frac{1}{2}$ chhoota) classic slip; area cm^2 se m^2 (10^{-4}) na badalne par ε_0 10^4 guna aa jaata hai.

Concept to recognise: An N-turn coil with the area in cm^2, $B = B_0\sin(\omega t)$ and 'average power': the time-varying-flux archetype; $NA\,dB/dt$ gives a sinusoidal emf and 'average' forces the rms step. *(Faraday emf from changing flux)*

Q47. Answer: 50

Alternating Current — Medium

Stem: A series LCR circuit with $R = 40\,\Omega$ and $L = \dfrac{0.8}{\pi}$ H is connected to a 220 V, 50 Hz AC supply. The current in the circuit lags behind the applied voltage and the power factor of the circuit is 0.8. The capacitive reactance of the circuit is _____ Ω.

1. $X_L = 2\pi f L = 2\pi(50)\dfrac{0.8}{\pi} = 80\,\Omega$ (220 V ki zarurat nahi padti).
2. $\cos\phi = 0.8 \Rightarrow \tan\phi = 3/4$; $|X_L - X_C| = R\tan\phi = 40 \times 0.75 = 30\,\Omega$ (ya $Z = R/\cos\phi = 50\,\Omega$ aur $\sqrt{Z^2 - R^2} = 30\,\Omega$).
3. Current lags, matlab circuit inductive, $X_L > X_C$: $X_C = 80 - 30 = 50\,\Omega$.

Final Answer: 50

Trap: 'Lags' ignore kar ke $X_C = X_L + 30 = 110\,\Omega$ (capacitive root) likh dena classic slip; 220 V redundant data hai, isse kuch nikalne ki koshish mat karo; Z mein square root bhoolna bhi common hai.

Concept to recognise: Series LCR with R, L, f, a stated power factor and the phrase 'current lags': the inverse power-factor archetype; $|X_L - X_C| = R\tan\phi$ has two roots and the lag picks one. *(Series LCR power factor from impedance)*

Q48. Answer: 16

Ray Optics — Medium

Stem: A thin convex lens of focal length 12 cm forms a virtual image three times the size of the object. The distance between the object and the image is _____ cm.

1. Convex lens se virtual image erect aur magnified hoti hai: $m = +3$. Lens ke liye $m = \dfrac{f}{f+u}$ (sign convention, $u < 0$).
2. $3 = \dfrac{12}{12+u} \Rightarrow 12 + u = 4 \Rightarrow u = -8$ cm; $v = mu = -24$ cm (image object wali side par).
3. Dono lens ke ek hi taraf: distance $= 24 - 8 = 16$ cm.

Final Answer: 16

Trap: Sign/direction slip: $m = -3$ (real image) lene par $u = -16$ cm, $v = +48$ cm aur distance 64 cm aata hai, alag integer; isliye virtual = erect = $m > 0$ ka decision hi asli question hai. $|u| + |v| = 32$ (side dekhe bina jodna) bhi galat.

Concept to recognise: 'Virtual image three times the size' from a convex lens with the object-image distance asked: the magnification-locates-object archetype; the sign of m is the whole decision and both lie on the same side. *(Magnification condition locates object or focus)*

Q49. Answer: 300

Ray Optics — Easy

Stem: A compound microscope has an objective of focal length 0.5 cm and an eyepiece of focal length 2.5 cm, with a tube length of 15 cm. In normal adjustment, the magnifying power of the microscope (take $D = 25$ cm) is _____.

1. Normal adjustment: $M = m_o \times m_e = \dfrac{L}{f_o} \times \dfrac{D}{f_e}$ (final image at infinity, isliye 1+ wala factor nahi).
2. $\dfrac{L}{f_o} = \dfrac{15}{0.5} = 30$; $\dfrac{D}{f_e} = \dfrac{25}{2.5} = 10$.
3. $M = 30 \times 10 = 300$.

Final Answer: 300

Trap: Formula misuse: final image at D wala factor $\left(1 + \dfrac{D}{f_e}\right) = 11$ laga dene par $M = 330$ (alag integer); tube length ko $f_o + f_e$ se badalna bhi common slip hai.

Concept to recognise: Objective and eyepiece focal lengths, a tube length and 'final image at infinity': the compound-microscope archetype; $(L/f_o)(D/f_e)$ with no 1+ because the image is at infinity. *(Compound microscope magnification)*

Q50. Answer: 61

Wave Optics — Easy

Stem: The objective of an astronomical telescope has a diameter of 11 cm. For light of wavelength 550 nm, the minimum angular separation between two distant point objects that the telescope can just resolve is _____ $\times 10^{-7}$ rad.

1. Rayleigh criterion: $\theta_{min} = \dfrac{1.22\lambda}{D}$.
2. $\lambda = 550 \times 10^{-9}$ m, $D = 0.11$ m: $\theta_{min} = \dfrac{1.22 \times 5.5 \times 10^{-7}}{0.11} = 6.1 \times 10^{-6}$ rad.
3. $= 61 \times 10^{-7}$ rad.

Final Answer: 61

Trap: Formula misuse: 1.22 chhod kar λ/D lene par 50; diameter ki jagah radius 5.5 cm lene par 122, dono alag integers; nm se m (10^{-9}) aur cm se m conversion sambhal kar.

Concept to recognise: Objective diameter, wavelength and 'just resolve' or Rayleigh: the resolving-limit archetype; $1.22\lambda/D$ with the answer scaled to 10^{-7} rad. *(Telescope resolving power Rayleigh)*

CHEMISTRY – SOLUTIONS

Q51. Answer: (2)

Haloalkanes & Haloarenes — Easy

Stem: Which of the following statements are correct? (A) Chlorobenzene is much less reactive than chloroethane towards nucleophilic substitution by aqueous KOH. (B) Among the isomeric alkyl bromides C_4H_9Br, 2-bromo-2-methylpropane has the highest boiling point. (C) 2-Bromobutane, on heating with alcoholic KOH, gives but-2-ene as the major product. (D) Ethyl fluoride can be prepared by heating ethyl chloride with NaF in dry acetone. (E) For a given alkyl group, the boiling points of the halides follow the order R−I > R−Br > R−Cl > R−F. Choose the correct answer from the options given below:

1. (A) Sahi: chlorobenzene mein C–Cl bond ko resonance se partial double-bond character milta hai aur carbon sp^2 hai, isliye aqueous KOH se S_N reaction bahut mushkil; chloroethane aasani se S_N2 deta hai.

2. (B) Galat: isomeric C_4H_9Br mein branching badhne par surface area ghatta hai, bp ghatta hai; 1-bromobutane sabse zyada, 2-bromo-2-methylpropane sabse kam.

3. (C) Sahi: alcoholic KOH se β-elimination, Saytzeff rule se zyada substituted alkene (but-2-ene) major.

4. (D) Galat: NaF se halogen exchange nahi hota; Finkelstein mein NaI/dry acetone (iodides) aur Swarts mein AgF, Hg_2F_2, CoF_2, SbF_3 (fluorides) lagte hain.

5. (E) Sahi: same R ke liye molar mass aur van der Waals forces ke saath bp badhta hai: R–I > R–Br > R–Cl > R–F.

Final Answer: (2) (A), (C) and (E) only

Trap: (1) = Finkelstein/Swarts confuse karke NaF-acetone ko fluoride route maan liya (D true); (3) = 'branching se bp badhta hai' wala ordering slip (B true); (4) = aryl halide ki inertness bhool kar chlorobenzene ko reactive maana (A false).

Concept to recognise: Five mixed haloalkane/haloarene facts (reactivity, bp trends, Saytzeff, halogen exchange) with 'only' combinations: the recall statement-set archetype; the two bp statements are the ordering traps. *(Haloalkane statement set: elimination, aryl inertness, SOCl2, trends)*

Q52. Answer: (1)

Hydrocarbons — Easy

Stem: The correct order of stability of the alkenes (P), (Q), (R) and (S) shown in the figure is

1. Alkene ki stability hyperconjugation se: C=C par jitne zyada alkyl groups (jitne zyada α-H), utna stable.

2. α-H ginte hain: (P) pent-1-ene $CH_2{=}CH{-}CH_2{-}$: 2; (Q) aur (R) pent-2-ene $CH_3{-}CH{=}CH{-}CH_2{-}$: 3 + 2 = 5; (S) $(CH_3)_2C{=}CH{-}CH_3$: 9. Isliye S sabse stable, P sabse kam.

3. (Q) aur (R) mein α-H barabar hain; cis isomer mein dono alkyl groups ek hi side hone se steric repulsion zyada, isliye trans (R) > cis (Q).

4. Order: S > R > Q > P.

Final Answer: (1) S > R > Q > P

Trap: (2) = cis ko trans se zyada stable maan liya; (3) = order ulta likh diya (alpha-H zyada = kam stable); (4) = sirf alpha-H gin kar cis/trans ka farq ignore kar diya (R = Q).

Concept to recognise: Four drawn alkenes differing in the number of alkyl groups on C=C plus a cis/trans pair: the hyperconjugation-count archetype; the cis/trans pair is where the order is decided. *(Alkene stability order (= Stability order of carbocations, carbanions and alkenes))*

Q53. Answer: (4)

Hydrocarbons + GOC — Medium

Stem: Given below are two statements: Statement I: 3-Methylbut-1-ene reacts with HBr in the absence of peroxide to give 2-bromo-3-methylbutane as the major product. Statement II: 3-Methylbut-1-ene, on treatment with Br_2 in CCl_4, then with excess $NaNH_2$ followed by water, and finally with $HgSO_4$ in dilute H_2SO_4, gives 3-methylbutan-2-one. In the light of the above statements, choose the most appropriate answer from the options given below:

1. Statement I (GOC): H^+ C-1 par judta hai (Markovnikov), C-2 par secondary carbocation $CH_3-\overset{+}{C}H-CH(CH_3)_2$ banta hai (4 α-H).

2. Paas ke C-3 se 1,2-hydride shift hokar tertiary carbocation $CH_3CH_2-\overset{+}{C}(CH_3)_2$ banta hai (8 α-H); hyperconjugation se $3^\circ > 2^\circ$, isliye rearrangement hota hai aur Br^- ke attack se major product 2-bromo-2-methylbutane milta hai. Statement I galat.

3. Statement II: Br_2/CCl_4 se 1,2-dibromo-3-methylbutane; excess $NaNH_2$ se do baar dehydrohalogenation aur terminal $\equiv$C–H ka deprotonation (sodium 3-methylbut-1-ynide); paani se 3-methylbut-1-yne.

4. $HgSO_4$/dil. H_2SO_4 hydration Markovnikov: enol tautomerise hokar 3-methylbutan-2-one (methyl ketone). Statement II sahi.

Final Answer: (4) Statement I is false but Statement II is true

Trap: (1) = sirf Markovnikov lagaya, carbocation rearrangement (1,2-H shift) bhool gaye, isliye Statement I ko sahi maana; (2) = rearrangement pakda lekin Kucherov hydration se aldehyde (3-methylbutanal) maan kar Statement II ko galat kaha; (3) = dono slips ek saath.

Concept to recognise: Two statements on one terminal alkene - its major HBr product and its Br2 / excess NaNH2 / HgSO4 roadmap: the alkyne-roadmap statement pair fused with a GOC carbocation statement on the same substrate. *(Alkyne synthesis and transformation roadmap)*

Q54. Answer: (3)

Coordination Compounds — Easy

Stem: The crystal field stabilisation energy (CFSE) of the tetrahedral complex $[CoCl_4]^{2-}$, expressed in terms of Δ_o, is (Given: $\Delta_t = \frac{4}{9}\Delta_o$)

1. Co^{2+}: $3d^7$. Tetrahedral complex hamesha high spin; splitting ulta hota hai: e (niche, $-0.6\Delta_t$) aur t_2 (upar, $+0.4\Delta_t$).

2. Filling: $e^4t_2^3$. CFSE $= 4(-0.6\Delta_t) + 3(+0.4\Delta_t) = -2.4\Delta_t + 1.2\Delta_t = -1.2\Delta_t$.

3. $\Delta_t = \frac{4}{9}\Delta_o$ rakhne par CFSE $= -1.2 \times \frac{4}{9}\Delta_o = -\frac{8}{15}\Delta_o$.

Final Answer: (3) $-\frac{8}{15}\Delta_o$

Trap: (1) = $-1.2\Delta_t$ nikal kar $\Delta_t \to \Delta_o$ conversion bhool gaye; (4) = tetrahedral ko octahedral pattern ($t_{2g}^5e_g^2$, $-0.4/+0.6$) se bhar diya: $-0.8\Delta_o$; (2) = wahi octahedral filling karke phir 4/9 se guna kar diya ($-16/45$).

Concept to recognise: A tetrahedral chloro complex with the CFSE demanded 'in terms of Delta_o' and the 4/9 relation given: the CFSE-computation archetype with the geometry conversion as the extra step. *(CFSE value computation and ordering)*

Q55. Answer: (2)

Chemical Bonding — Medium

Stem: Among N_2, O_2, F_2 and N_2^-, the species for which the removal of one electron decreases the bond order but increases the number of unpaired electrons is

1. N_2 (14 e): $\sigma 2s^2\,\sigma^*2s^2\,\pi 2p^4\,\sigma 2p_z^2$, BO = 3, unpaired 0. Electron bonding $\sigma 2p_z$ se nikalta hai: N_2^+ BO = 2.5, unpaired 1. Dono sharten poori.

2. O_2 (16 e): π^*2p^2, BO = 2, unpaired 2; electron antibonding π^* se nikalta hai: O_2^+ BO = 2.5 (badha), unpaired 1 (ghata).

3. F_2 (18 e): BO = 1, unpaired 0; F_2^+ (π^*2p^3) BO = 1.5 (badha), unpaired 1.

4. N_2^- (15 e): π^*2p^1, BO = 2.5, unpaired 1; electron π^* se nikal kar N_2 banta hai: BO = 3 (badha), unpaired 0 (ghata).

Final Answer: (2) N_2

Trap: Main slip = antibonding ko ignore karke 'electron nikla to BO ghata' maan lena: is slip mein unpaired-count wali shart O_2 ($2 \to 1$) aur N_2^- ($1 \to 0$) ko khud reject kar deti hai, sirf F_2 ($0 \to 1$) bachta hai, isliye yeh slip exactly (4) par le jaata hai; (3) = N_2^- ka -1 charge ignore karke use N_2 jaisa maan liya; (1) = O_2 ka electron bonding π se hataya samajh kar unpaired $2 \to 3$ gin liya.

Concept to recognise: 'Removal of one electron' with both a bond-order and an unpaired-electron condition over N2, O2, F2 and N2^-: the MOT-configuration archetype asked as cation-vs-parent comparison; the identity of each species' HOMO (bonding or antibonding) decides. *(MOT bond order and magnetism)*

Q56. Answer: (4)

GOC — Easy

Stem: The structures (A), (B), (C) and (D) shown in the figure are resonating structures of prop-2-enal ($CH_2{=}CH{-}CHO$). The least stable resonating structure is

1. Resonating structure ki stability ke niyam: zyada covalent bonds, har atom ka octet poora, charge separation kam, aur negative charge zyada electronegative atom par.

2. (A): koi charge nahi, sab octet poore: sabse stable.

3. (B) aur (C): charge separated hain, carbon par + (sextet) lekin negative charge electronegative O par: theek-thaak contributors.

4. (D): positive charge oxygen par (zyada electronegative atom) aur us O ke paas sirf 6 electron; negative charge carbon par: sabse kam stable.

Final Answer: (4) (D)

Trap: (3) = 'charges jitne door, utna unstable' wala formula-misuse (C ko least maan liya); (2) = carbonyl ke paas wale C^+ ko sabse unstable samjha; (1) = 'least' ki jagah 'most' stable chun liya.

Concept to recognise: Four drawn charge-separated forms of a conjugated enal with 'least stable' asked: the resonance-ranking archetype; look for the structure whose charges sit on the wrong atoms. *(Resonance contributor stability ranking)*

Q57. Answer: (1)

Chemical Kinetics — Medium

Stem: The gas-phase decomposition A(g) $\rightarrow$ 2B(g) + C(g) is a first-order reaction. Starting with pure A at an initial pressure of 400 mm Hg in a closed vessel at constant temperature, the total pressure becomes 600 mm Hg after 10 minutes. The rate constant of the reaction is (Given: $\log 2 = 0.30$, $\log 3 = 0.48$)

1. Pressure table: A: $400 - x$, B: $2x$, C: x; total $= 400 + 2x$.
2. $400 + 2x = 600 \Rightarrow x = 100$ mm Hg, isliye $P_A = 300$ mm Hg.
3. $k = \frac{2.303}{t} \log \frac{P_0}{P_A} = \frac{2.303}{10} \log \frac{400}{300} = 0.2303\,(2\log 2 - \log 3) = 0.2303 \times 0.12$.
4. $k = 2.76 \times 10^{-2}$ min^{-1}.

Final Answer: (1) 2.76×10^{-2} **min**$^{-1}$

Trap: (4) = total pressure $400 + x$ le liya (coefficient 2 bhool gaye): $x = 200$, $\log 2$; (3) = $\log(P_t/P_0) = \log 1.5$ laga diya (P_0 vs P_t confusion); (2) = total $400 + 3x$ (coefficients ka sum A ke liye bhi laga diya): $\log 1.2$.

Concept to recognise: First-order gas decomposition with initial and total pressure at one time and a 1 -> 3 stoichiometry: the pressure-method archetype; the coefficient on x in the total-pressure relation is the whole question. *(First-order gas-phase pressure method)*

Q58. Answer: (3)

Purification & Characterisation — Medium

Stem: In the Liebig combustion analysis of 0.25 g of an organic compound, the increase in the mass of the U-tube containing anhydrous $CaCl_2$ was 225 mg and that of the bulb containing KOH solution was 660 mg. The mass of oxygen present in the sample is

1. $CaCl_2$ tube paani sokhta hai (225 mg H_2O), KOH bulb CO_2 (660 mg).
2. Carbon: $\frac{12}{44} \times 660 = 180$ mg. Hydrogen: $\frac{2}{18} \times 225 = 25$ mg.
3. Sample 0.25 g = 250 mg; oxygen (by difference) $= 250 - 180 - 25 = 45$ mg.

Final Answer: (3) 45 mg

Trap: (1) = hydrogen bhool kar $250 - 180 = 70$ mg; (2) = H ke liye $\frac{1}{18}$ (12.5 mg) laga diya: 57.5 mg; (4) = gram mein 0.045 g nikal kar mg mein badalte waqt power of ten ka slip (4.5 mg).

Concept to recognise: CaCl2 tube and KOH bulb mass gains with the sample in g and the gains in mg: the Liebig C/H estimation; the asked oxygen comes by difference after converting both products. *(Liebig combustion carbon hydrogen percent)*

Q59. Answer: (2)

Biomolecules — Easy

Stem: Consider the following amino acids: Valine, Glycine, Lysine, Methionine, Proline The number of essential amino acids among the above is :

1. Essential amino acids wo hain jo sharir khud nahi bana sakta, diet se lene padte hain (NCERT table mein asterisk wale): Val, Leu, Ile, Arg, Lys, Thr, Met, Phe, Trp, His.
2. List mein essential: Valine, Lysine, Methionine = 3; Glycine aur Proline non-essential.

Final Answer: (2) 3

Trap: (1) = ulta gin liya (non-essential ki ginti 2: direction slip); (3) = proline (ya glycine) ko essential maan liya: 4; (4) = list ke sabhi amino acids ko essential maan liya: 5.

Concept to recognise: A five-name amino-acid list followed by 'the number of essential amino acids': the NCERT-table recall count. *(Amino acid codes, class and side-chain recognition)*

Q60. Answer: (1)

d & f Block — Easy

Stem: Which of the following statements about lanthanoids are correct? (A) Ce^{4+} is a strong oxidising agent in aqueous solution and gets reduced to Ce^{3+}. (B) Eu^{2+} is a strong reducing agent and changes to the +3 state. (C) Tb^{4+} and Gd^{3+} have the same number of 4f electrons. (D) Tb^{3+} has a half-filled 4f subshell. (E) As a consequence of lanthanoid contraction, the atomic radii of Zr and Hf are almost identical. Choose the correct answer from the options given below:

1. (A) Sahi: Ce^{4+} ($E^\circ = +1.74$ V) strong oxidant hai, common +3 state mein wapas aata hai. (B) Sahi: Eu^{2+} ($4f^7$) strong reducing agent, +3 mein jaata hai.
2. Configurations: $Ln^{3+} = 4f^{Z-57}$. Gd^{3+} ($Z = 64$): $4f^7$; Tb ($Z = 65$): $Tb^{3+} = 4f^8$, $Tb^{4+} = 4f^7$.
3. (C) Sahi: dono $4f^7$. (D) Galat: Tb^{3+} $4f^8$ hai, half-filled nahi (isi liye Tb(IV) banta hai).
4. (E) Sahi: lanthanoid contraction se 5d series ke radii 4d ke lagbhag barabar: Zr 160 pm, Hf 159 pm.

Final Answer: (1) (A), (B), (C) and (E) only

Trap: (2) = Tb^{3+} ko $4f^7$ maan liya aur (C) chhod diya; (3) = Ce^{4+} ko reducing samjha (direction slip); (4) = Eu^{2+} ko oxidising aur Tb^{3+} ko half-filled maan liya.

Concept to recognise: Five lanthanoid statements mixing +2/+4 redox nature, 4f counts and the Zr/Hf radius pair: the f-configuration/redox statement set; the Tb3+/Tb4+ count is the decider. *(Lanthanoid ion f-configuration and redox nature)*

Q61. Answer: (4)

p-Block + d & f Block — Easy

Stem: Which of the following statements is correct?

1. (1) Galat: F_2 mein chhote bond ki wajah se lone pairs ka repulsion zyada; BDE order $Cl_2 > Br_2 > F_2 > I_2$.
2. (3) Galat: F zyada electronegative hai, isliye OF_2 mein O ka oxidation state +2 hai.
3. (2) aur (4) d-block se tay hote hain: NCERT ke anusaar high oxidation states ko stabilise karne mein oxygen fluorine se behtar hai, kyunki oxygen metal ke saath multiple bonds bana sakta hai. Isliye Mn ka highest fluoride MnF_4 (+4) hai jabki highest oxide Mn_2O_7 (+7). (2) galat, (4) sahi.

Final Answer: (4) The highest fluoride of manganese is MnF_4, whereas its highest oxide is Mn_2O_7.

Trap: (2) = p-block wali soch 'fluorine sabse high oxidation state laata hai' (IF_7, SF_6 jaise) ko Mn par laga diya: d-block exception (oxygen multiple bonding) bhool gaye; (1) = F_2 bond anomaly bhool kar 'sabse electronegative to sabse strong bond' maana; (3) = oxygen ko hamesha -2 maan liya.

Concept to recognise: Four single statements on F2 bond enthalpy, the oxidation state of O in OF2 and how O and F stabilise manganese's highest oxidation state: the Group 16/17 anomaly set fused with d-block oxidation-state stability. *(Group 16/17 anomaly and oxidising trend statements)*

Q62. Answer: (3)

Chemical Thermodynamics + Kinetic Theory — Medium

Stem: At 298 K, the correct order of molar heat capacities at constant pressure ($C_{p,m}$) of $H_2O(l)$, $N_2(g)$, Al(s) and Ar(g) is

1. Liquid water: specific heat 4.18 J g^{-1} K^{-1}, isliye $C_{p,m} = 4.18 \times 18 \approx 75$ J K^{-1} mol^{-1} (sabse zyada).
2. Gases (kinetic theory, equipartition): monatomic Ar $C_{V,m} = \frac{3}{2}R$, $C_{p,m} = C_{V,m} + R = \frac{5}{2}R \approx 20.8$; diatomic N_2 (3 translational + 2 rotational) $C_{V,m} = \frac{5}{2}R$, $C_{p,m} = \frac{7}{2}R \approx 29.1$ J K^{-1} mol^{-1}.
3. Metal Al (Dulong–Petit): $C_m \approx 3R \approx 24.9$ J K^{-1} mol^{-1}, jo $\frac{7}{2}R$ se kam aur $\frac{5}{2}R$ se zyada hai.
4. Order: $H_2O(l)$ (≈ 75) > $N_2(g)$ ($\frac{7}{2}R$) > Al(s) ($3R$) > Ar(g) ($\frac{5}{2}R$).

Final Answer: (3) $H_2O(l) > N_2(g) > Al(s) > Ar(g)$

Trap: (1) = gases ke liye C_V ($\frac{5}{2}R$ aur $\frac{3}{2}R$) le liya, to N_2 (20.8) Al ($3R$) se neeche chala jaata hai: C_p vs C_V ka direction slip; (2) = 'gases ki heat capacity sabse zyada' wali galatfehmi; (4) = Dulong–Petit $3R$ ko sabse bada maan kar solid ko top par rakh diya.

Concept to recognise: A liquid, a metal, a diatomic and a monatomic gas to be ordered by molar heat capacity 'at constant pressure': the Dulong-Petit vs equipartition archetype; the C_p (not C_V) values of the gases decide the metal-vs-N2 pair. *(Molar heat capacity across phases order)*

Q63. Answer: (1)

Ionic Equilibrium — Easy

Stem: The pH of a 0.1 M aqueous solution of sodium cyanide at 298 K is (Given: K_a of HCN = 1.0×10^{-10})

1. NaCN weak acid (HCN) + strong base (NaOH) ka salt hai; CN^- ka anionic hydrolysis solution ko basic banata hai: $CN^- + H_2O \rightleftharpoons HCN + OH^-$, $K_h = K_w/K_a$.
2. $pK_a = -\log(1.0 \times 10^{-10}) = 10$.
3. $\text{pH} = 7 + \frac{1}{2}pK_a + \frac{1}{2}\log c = 7 + 5 + \frac{1}{2}(-1) = 11.5$.

Final Answer: (1) 11.5

Trap: (2) = cationic hydrolysis wala formula ($7 - \frac{1}{2}pK_a - \frac{1}{2}\log c$) ya pOH ko pH likh diya: 2.5; (3) = $\frac{1}{2}\log c$ term chhod di: 12.0; (4) = $\log c$ par $\frac{1}{2}$ nahi lagaya: 11.0.

Concept to recognise: A 0.1 M solution of the sodium salt of a weak acid with only Ka given and pH asked: the salt-hydrolysis archetype (anionic hydrolysis); the sign pattern of the pH formula is the whole test. *(legacy archetype (writer supplies))*

Q64. Answer: (2)

Coordination Compounds — Medium

Stem: The IUPAC name of the complex $K[Cr(en)(C_2O_4)_2]$ (en = ethane-1,2-diamine) is

1. Ek K^+ hai, isliye complex ion ka charge -1. Ligands: en (neutral, bidentate), do oxalato $C_2O_4^{2-}$ (-2 each): $x + 0 - 4 = -1 \Rightarrow Cr = +3$.
2. Ligands alphabetical order mein, numerical prefix (di) ko ignore karke: ethane-1,2-diamine (e) pehle, oxalato (o) baad mein; en ka naam bracket mein likha jaata hai.
3. Complex anionic hai, isliye metal ka naam 'chromate' aur oxidation state Roman mein: chromate(III); cation potassium pehle.
4. Naam: potassium (ethane-1,2-diamine)dioxalatochromate(III).

Final Answer: (2) Potassium (ethane-1,2-diamine)dioxalatochromate(III)

Trap: (1) = 'dioxalato' ke 'd' ko alphabetical order mein gin liya (prefix ignore karna tha); (3) = K^+ bhool kar complex ko neutral maana: $x - 4 = 0$, Cr(IV); (4) = anionic complex ke liye '-ate' nahi lagaya.

Concept to recognise: A potassium salt of a chromium complex carrying en and two oxalato ligands with four names differing in ligand order, oxidation number or the -ium/-ate ending: the nomenclature archetype; apply the alphabetical, charge-balance and anion-ending rules in turn. *(legacy archetype (writer supplies))*

Q65. Answer: (4)

Solutions + Purification & Characterisation — Easy

Stem: Given below are two statements: Statement I: Steam distillation is used to separate substances which are steam volatile and miscible with water. Statement II: Solutions showing a large positive deviation from Raoult's law form minimum boiling azeotropes. In the light of the above statements, choose the most appropriate answer from the options given below:

1. Statement I (purification): steam distillation un substances ke liye hai jo steam volatile hon aur paani mein immiscible hon (jaise aniline–water); miscible substance paani se alag hi nahi hoga. Statement I galat.

2. Statement II (solutions): large positive deviation mein A–B interactions A–A aur B–B se kamzor hote hain, vapour pressure zyada aur ek khaas composition par minimum boiling azeotrope banta hai (jaise ethanol–water). Statement II sahi.

Final Answer: (4) Statement I is false but Statement II is true

Trap: (1) = 'miscible' shabd par dhyan nahi diya (steam distillation ki shart immiscible hai); (2) = positive deviation ko maximum boiling azeotrope se jod diya; (3) = dono slips ek saath.

Concept to recognise: A distillation-technique statement paired with an NCERT sentence linking the deviation sign to the azeotrope type: the azeotrope statement pair fused with purification techniques; each clause is checked against NCERT wording. *(Non-ideal solutions: deviations and azeotropes)*

Q66. Answer: (3)

Atomic Structure — Medium

Stem: The wavelength of the series limit of the Balmer series of He^+ is (Given: $R_H = 1.1 \times 10^7\ m^{-1}$)

1. Balmer series: $n_1 = 2$; series limit (shortest λ) ke liye $n_2 \to \infty$.

2. $\bar{\nu} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = 1.1 \times 10^7 \times 4 \times \frac{1}{4} = 1.1 \times 10^7\ m^{-1}$ (He^+ ke liye $Z = 2$).

3. $\lambda = \frac{1}{\bar{\nu}} = 9.09 \times 10^{-8}$ m $= 90.9$ nm.

Final Answer: (3) 90.9 nm

Trap: (1) = Z^2 bhool gaye (hydrogen ki Balmer limit 364 nm); (2) = Z^2 ki jagah Z laga diya; (4) = 9.09×10^{-8} m ko nm mein badalte waqt power of ten ka slip (9.09 nm).

Concept to recognise: 'Series limit of the Balmer series' for a one-electron ion with R_H in mˆ-1 and options in nm: the Rydberg archetype; n2 -> infinity, Zˆ2 and the m-to-nm conversion are the three checkpoints. *(Rydberg spectral line energy comparison)*

Q67. Answer: (1)

Alcohols Phenols & Ethers — Medium

Stem: The number of structures among (A)-(H) shown in the figure that are secondary alcohols and are also chiral is

1. Pehle degree: OH wale carbon (carbinol carbon) par kitne carbon jude hain. (B) CH_2OH: 1°; (C) teen carbon: 3°; baaki (A), (D), (E), (F), (G), (H) sab 2° (chhah).

2. Ab chirality: carbinol carbon par chaar alag groups chahiye.

3. (A) cyclohexanol: dono ring arms same (CH_2CH_2-): achiral. (G) pentan-3-ol: do ethyl groups: achiral.

4. (D) C_6H_5, CH_3, H, OH: chiral. (E) CH_3, $CH(CH_3)_2$, H, OH: chiral. (F) vinyl, CH_3, H, OH: chiral. (H) ring mein ek taraf CH=, doosri taraf CH_2: chaar alag groups, chiral.

5. Secondary aur chiral: D, E, F, H = 4.

Final Answer: (1) 4

Trap: (2) = sirf secondary gin liye, chirality check nahi ki (6); (3) = ring wale (H) ko achiral maan kar chhod diya (3); (4) = pentan-3-ol ya cyclohexanol ko bhi chiral gin liya (5).

Concept to recognise: Eight drawn alcohols (cyclic, allylic, benzylic, branched) with 'secondary alcohols and also chiral' asked: the 1/2/3-degree count archetype with a second filter; classify by the carbinol carbon, then test

it for four different groups. *(Classify alcohols 1/2/3 degree count)*

Q68. Answer: (4)

Electrochemistry — Easy

Stem: According to the electrochemical theory of rusting of iron, the reaction occurring at the cathodic area on the surface of iron is

1. Rusting ek electrochemical cell jaisa hai: metal ka ek spot anode banta hai jahan $Fe \rightarrow Fe^{2+} + 2e^-$ ($E^\circ = -0.44$ V).

2. Electrons metal se hokar doosre spot (cathode) par jaate hain, jahan ghule hue CO_2 se mile H^+ ki maujoodgi mein oxygen reduce hota hai: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$ ($E^\circ = 1.23$ V).

3. Baad mein Fe^{2+} hawa ke oxygen se Fe^{3+} mein oxidise hokar hydrated ferric oxide (rust, $Fe_2O_3 \cdot xH_2O$) banata hai; yeh cathodic half-reaction nahi hai.

Final Answer: (4) $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$

Trap: (1) = anodic reaction ko cathodic samajh liya; (2) = acidic medium mein corrosion tez hota hai sun kar H_2 evolution ko cathode par maan liya; (3) = rust banne wale baad ke oxidation step ko cathode par rakh diya.

Concept to recognise: 'Electrochemical theory of rusting' together with 'cathodic area' in the stem: the NCERT corrosion-mechanism recall archetype. *(Corrosion mechanism statements)*

Q69. Answer: (2)

Solutions — Medium

Stem: When 4.5 g of a non-volatile, non-electrolyte solute is dissolved in 147 g of acetic acid, the boiling point of the solvent is raised by 1.02 K. The relative lowering of vapour pressure of this solution is (Given: K_b of acetic acid = 3.0 K kg mol^{-1}, molar mass of acetic acid = 60 g mol^{-1})

1. $\Delta T_b = K_b m \Rightarrow m = \frac{1.02}{3.0} = 0.34$ mol kg^{-1}.

2. Solute ke moles $n_s = 0.34 \times 0.147 = 0.05$ mol (molar mass $= 4.5/0.05 = 90$ g mol^{-1}).

3. Solvent ke moles $n_{solv} = \frac{147}{60} = 2.45$ mol.

4. Relative lowering $= \frac{p^\circ - p}{p^\circ} = x_{solute} = \frac{0.05}{0.05+2.45} = \frac{0.05}{2.50} = 0.02$.

Final Answer: (2) 0.02

Trap: (1) = molality ko hi mole fraction maan liya (0.34); (3) = $m = K_b/\Delta T_b$ ulta kar diya: $n_s = 0.43$, $x = 0.15$; (4) = solvent ka mole fraction $(1 - 0.02)$ likh diya.

Concept to recognise: Boiling-point rise data with the solvent's molar mass given and 'relative lowering of vapour pressure' asked: the linked colligative archetype; moles from dTb, then an exact mole fraction (solvent moles needed). *(Ebullioscopy molar mass and linked RLVP)*

Q70. Answer: (3)

Amines — Medium

Stem: Match List-I with List-II.

List-I (Amine)	List-II (Observation)
(A) $C_6H_5NHCH_3$	(I) Liberates N_2 gas quantitatively with $NaNO_2$ and dil. HCl at 273–278 K
(B) $C_6H_5N(CH_3)_2$	(II) Does not react with benzenesulphonyl chloride
(C) $C_6H_5NH_2$	(III) Forms a diazonium salt, stable at 273–278 K, with $NaNO_2$ and dil. HCl
(D) $C_6H_5CH_2NH_2$	(IV) Gives with benzenesulphonyl chloride a product insoluble in aqueous KOH

Choose the correct answer from the options given below:

1. Pehle har amine ko classify karo: (A) N-methylaniline 2°; (B) N,N-dimethylaniline 3°; (C) aniline 1° aromatic (N seedha ring par); (D) benzylamine 1° aliphatic (N, CH_2 par hai, ring par nahi).

2. Hinsberg's reagent ($C_6H_5SO_2Cl$): 2° amine ka sulphonamide N par H nahi rakhta, alkali mein insoluble: (A)-(IV). 3° amine react hi nahi karta: (B)-(II).

3. Nitrous acid: aromatic 1° amine 273–278 K par diazonium salt deta hai jo us temperature par stable hai: (C)-(III).

4. Aliphatic 1° amine ka diazonium salt unstable hai, N_2 quantitatively nikalta hai (alcohol banta hai): (D)-(I).

Final Answer: (3) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

Trap: (1) = Hinsberg mein 2° aur 3° ka behaviour ulta (2-deg ko 'no reaction' maana); (2) = benzylamine mein benzene ring dekh kar use aromatic amine maan liya, aniline aur benzylamine ke nitrous-acid results badal diye; (4) = dono near-miss slips ek saath.

Concept to recognise: Four amines of different degree and type matched to benzenesulphonyl-chloride and nitrous-acid observations: the amine test-chemistry match archetype; classify each amine (degree; N on the ring or not), then apply each test's scope. *(Named amine reactions: scope, reagent match, test chemistry)*

Q71. Answer: 39

Chemical Thermodynamics — Medium

Stem: For a reaction, the equilibrium constant is $K = 1.0 \times 10^{-2}$ at 500 K and the standard entropy change is $\Delta S^\circ = +40.0$ J K^{-1} mol^{-1}. The standard enthalpy change ΔH° of the reaction at 500 K is _____ kJ mol^{-1} (nearest integer). (Given: $R = 8.314$ J K^{-1} mol^{-1}, $\ln 10 = 2.303$)

1. $\Delta G^\circ = -2.303\,RT \log K = -2.303 \times 8.314 \times 500 \times \log(10^{-2}) = -2.303 \times 8.314 \times 500 \times (-2) = +19147$ J mol^{-1} ($K < 1$, isliye ΔG° positive).

2. $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ \Rightarrow \Delta H^\circ = \Delta G^\circ + T\Delta S^\circ = 19147 + 500 \times 40.0 = 39147$ J mol^{-1}.

3. $\Delta H^\circ \approx 39.1$ kJ mol$^{-1} \approx 39$.

Final Answer: 39

Trap: Sign slip: $\log K = -2$ ka minus kho kar $\Delta G^\circ = -19147$ J le liya to $\Delta H^\circ = -19147 + 20000 = 853$ J ≈ 1 kJ (alag integer); unit slip: ΔS° ko kJ maan kar $T\Delta S^\circ = 20$ J lene par 19 kJ.

Concept to recognise: K at one temperature plus dS given and dH asked: the dG-dH-dS-K linkage; the sign of log K (K < 1) fixes the sign of dG before Gibbs-Helmholtz is applied. *(Gibbs energy from dH, dS and K)*

Q72. Answer: 175

Chemical Equilibrium + Chemical Thermodynamics — Medium

Stem: At 350 K, pure N_2O_4 is allowed to attain the equilibrium $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ in a closed vessel. At equilibrium, the total pressure is 3 atm and N_2O_4 is 50% dissociated. If the standard enthalpy change of the reaction is $\Delta_r H^\circ = +57.2$ kJ mol^{-1}, the standard entropy change $\Delta_r S^\circ$ of the reaction at 350 K is _____ J K^{-1} mol^{-1} (nearest integer). (Given: $R = 8.314$ J K^{-1} mol^{-1}, $\log 2 = 0.30$, $\ln 10 = 2.303$. K_p is expressed in atm.)

1. ICE table (1 mol N_2O_4, $\alpha = 0.5$): N_2O_4 0.5, NO_2 1.0, total $1 + \alpha = 1.5$ mol; $P = 3$ atm se $p_{NO_2} = 2$ atm, $p_{N_2O_4} = 1$ atm.

2. $K_p = \frac{p^2_{NO_2}}{p_{N_2O_4}} = 4$ atm; $\Delta_r G^\circ = -2.303\,RT \log K_p$.

3. Thermodynamics: $\Delta_r G^\circ = \Delta_r H^\circ - T\Delta_r S^\circ \Rightarrow \Delta_r S^\circ = \frac{\Delta_r H^\circ}{T} + 2.303\,R \log K_p$.

4. $\Delta_r S^\circ = \frac{57200}{350} + 2.303 \times 8.314 \times 0.60 = 163.43 + 11.49 = 174.9 \approx 175$ J K^{-1} mol^{-1}.

Final Answer: 175

Trap: $\Delta_r G^\circ$ ka sign ulta (+4021 J) lene par 152; $\Delta_r H^\circ$ ko kJ mein hi rakhne par 12; total moles $(1 + \alpha)$ chhodne par $K_p = 6$ aur 178; 2.303 bhoolne par 168: sab alag integers.

Concept to recognise: Degree of dissociation and total pressure for a 1 -> 2 gas equilibrium plus the standard enthalpy, with the standard entropy asked: the ICE-table archetype feeding dG0 = -2.303RT log K and then dG0 = dH0 - T dS0; the (1+alpha) in the mole-fraction denominators is the hidden condition. *(ICE table: dissociation degree, Kp and partial pressures)*

Q73. Answer: 547

Periodic Properties + Mole Concept — Easy

Stem: The energy required to convert 6.0 mg of gaseous magnesium atoms completely into $Mg^{2+}(g)$ ions is _____ J. (Given: $\Delta_i H_1$ and $\Delta_i H_2$ of Mg are 737 and 1451 kJ mol^{-1} respectively; molar mass of Mg = 24 g mol^{-1})

1. Mole concept: $n(\text{Mg}) = \frac{6.0\times10^{-3}\ \text{g}}{24\ \text{g mol}^{-1}} = 2.5 \times 10^{-4}$ mol.

2. Mg(g) se $Mg^{2+}(g)$ ke liye dono successive ionisation enthalpies chahiye: $\Delta_i H_1 + \Delta_i H_2 = 737 + 1451 = 2188$ kJ mol^{-1}.

3. Energy $= 2.5 \times 10^{-4} \times 2188 = 0.547$ kJ $= 547$ J.

Final Answer: 547

Trap: Sirf $\Delta_i H_1$ lene par 184 J; mg ko g maan lene par 5.47×10^5 J; kJ ko J mein na badalne par 0.547 likh denge (integer nahi).

Concept to recognise: A milligram mass of a gaseous metal to be taken to the +2 ion with two ionisation enthalpies given: the ionisation-energy-for-a-sample archetype; sum both IEs and convert mg to mol. *(Ionisation energy for a sample mass)*

Q74. Answer: 9

Aldehydes Ketones & Carboxylic Acids + Amines — Easy

Stem: 14.8 g of propanoic acid is converted into its acid chloride with $SOCl_2$ and then into the amide B with excess NH_3. B on heating with Br_2 and aqueous NaOH gives the amine C. Assuming quantitative conversion in each step, the mass of C obtained is _____ g. (Given: atomic masses C = 12, H = 1, N = 14, O = 16)

1. Propanoic acid CH_3CH_2COOH, $M = 74$ g mol^{-1}: $n = \frac{14.8}{74} = 0.20$ mol.

2. $SOCl_2$ se propanoyl chloride, NH_3 se B = propanamide $CH_3CH_2CONH_2$ (0.20 mol).

3. Hoffmann bromamide degradation (Amines): Br_2 + NaOH se amide ka ek carbon kam hokar C = ethanamine $CH_3CH_2NH_2$ (0.20 mol).

4. $M(C_2H_7N) = 24 + 7 + 14 = 45$ g mol^{-1}; mass $= 0.20 \times 45 = 9.0$ g.

Final Answer: 9

Trap: Counting slip: Hoffmann mein carbon loss bhool kar propan-1-amine ($M = 59$) lene par $11.8 \approx 12$ g (alag integer); amide par ruk jaane par $0.2 \times 73 = 14.6 \approx 15$ g.

Concept to recognise: An acid taken to its amide and then Br2/NaOH with the amine's mass asked from a given mass of acid: the acid-derivative roadmap; the Hofmann step shortens the chain before the molar mass is taken. *(Carboxylic acid and derivative sequences)*

Q75. Answer: 50

Amines + Purification & Characterisation — Medium

Stem: Acetanilide on treatment with Br_2 in acetic acid gives the major product Q, which on hydrolysis with aqueous NaOH gives R. In Kjeldahl's method, the ammonia evolved from 1.72 g of R is completely neutralised by V mL of 0.1 M H_2SO_4. The value of V is _____. (Given: atomic masses C = 12, H = 1, N = 14, O = 16, Br = 80)

1. Acetanilide mein $NHCOCH_3$ group ortho/para-directing hai lekin bulky, isliye Br_2/CH_3COOH se Q = p-bromoacetanilide (major).

2. Aqueous NaOH se amide hydrolysis: R = p-bromoaniline C_6H_6BrN, $M = 72 + 6 + 80 + 14 = 172$ g mol^{-1}.

3. Kjeldahl (Purification & Characterisation): har N atom NH_3 banata hai; $n(NH_3) = \frac{1.72}{172} = 0.01$ mol.

4. $2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$: $n(H_2SO_4) = 0.005$ mol; $V = \frac{0.005}{0.1} = 0.05$ L $= 50$ mL.

Final Answer: 50

Trap: Hydrolysis step bhool kar R ko p-bromoacetanilide ($M = 214$) maanne par $V \approx 40$ mL; $NH_3 : H_2SO_4$ ko 1 : 1 lene par 100 mL; acetanilide ($M = 135$) par Kjeldahl lagane par 64 mL.

Concept to recognise: Bromination then hydrolysis of acetanilide followed by a Kjeldahl acid volume on the prod-

uct: the aniline-derivative calculation archetype; the product's molar mass after hydrolysis and the 2:1 NH3:H2SO4 ratio carry the answer. *(Aniline-derivative transformation with mass or %N calculation)*